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Drach et al.

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(54) **SYSTEMS AND METHODS FOR VEHICLE SUSPENSIONS**

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B60G 17/015 (2006.01)
(Continued)

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(2013.01); **B62D 7/15** (2013.01);
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(Continued)

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Primary Examiner — Paul N Dickson

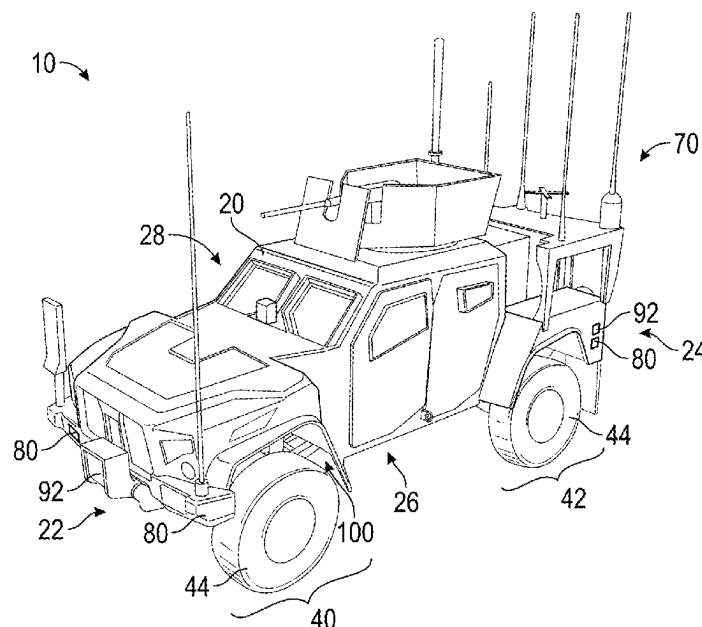
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(57) **ABSTRACT**

A suspension element includes a housing, a first joint, and a second joint. The housing is configured to couple a tractive element assembly to a vehicle. The housing has a first end configured to engage a portion of the vehicle and a second end configured to interface with the tractive element assembly. The first joint includes a first actuator and a first resilient member. The first actuator is configured to facilitate linear extension and retraction of the suspension element. The second joint includes a second actuator and a second resilient member. The second actuator is configured to facilitate rotational movement of the suspension element. The first resilient member and the second resilient member are configured to support a static load of the vehicle.

20 Claims, 24 Drawing Sheets



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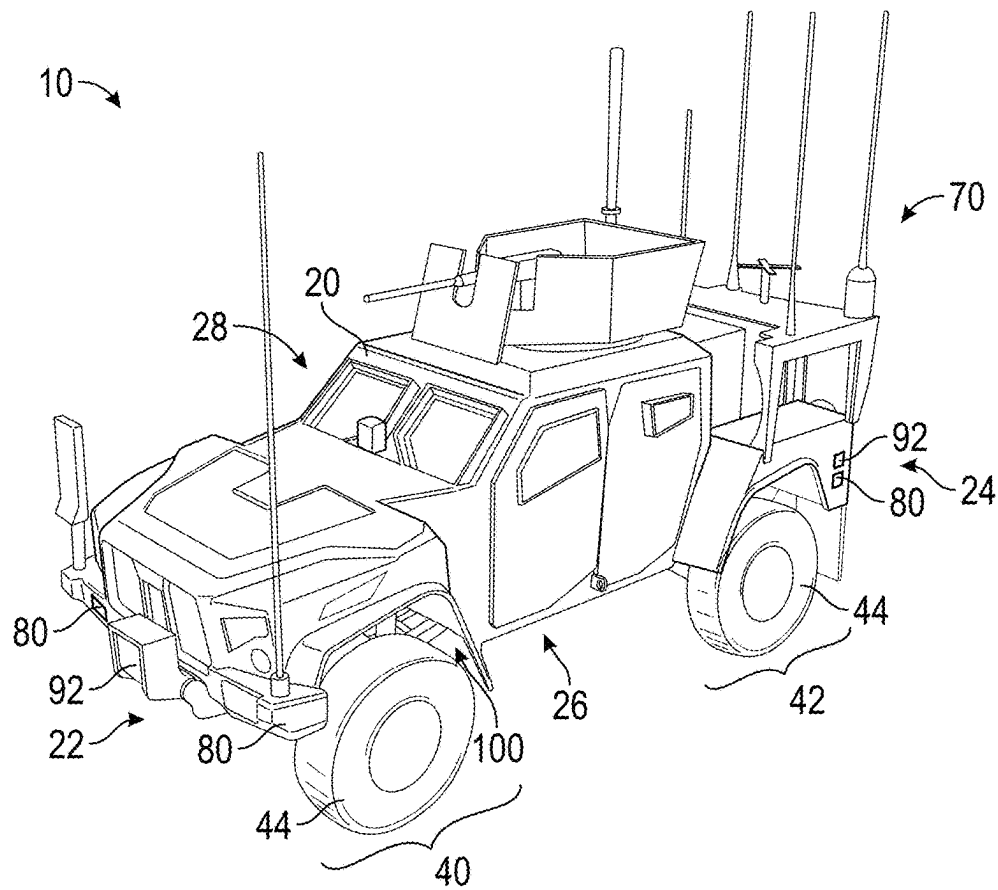


FIG. 1

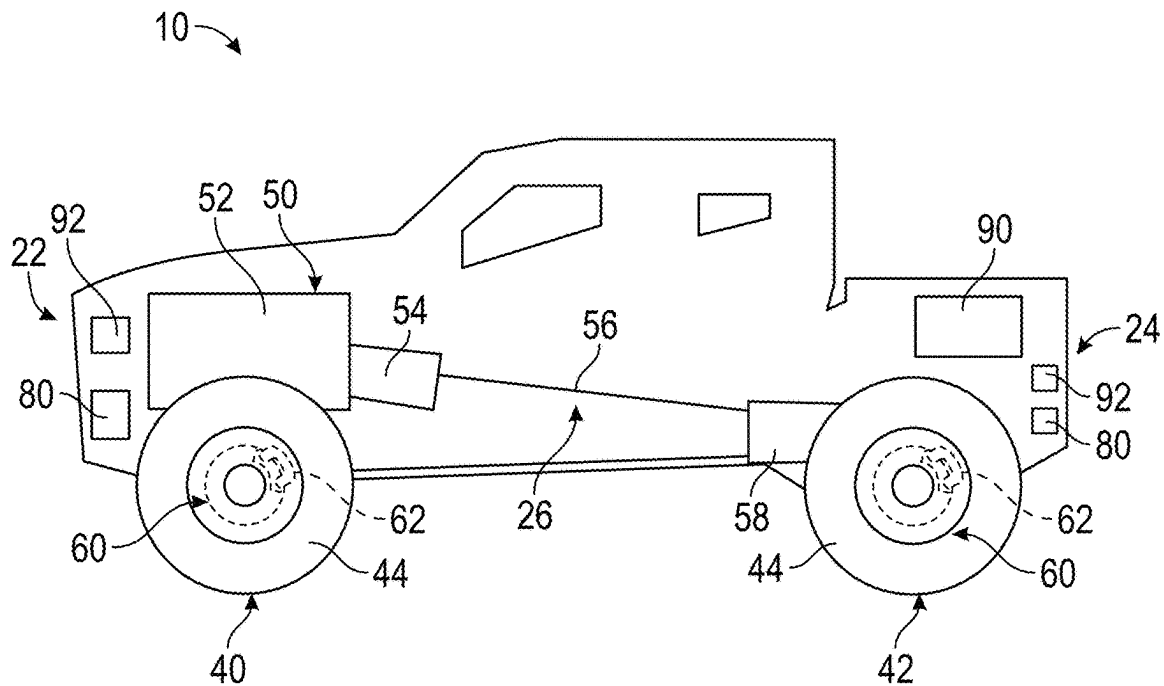


FIG. 2

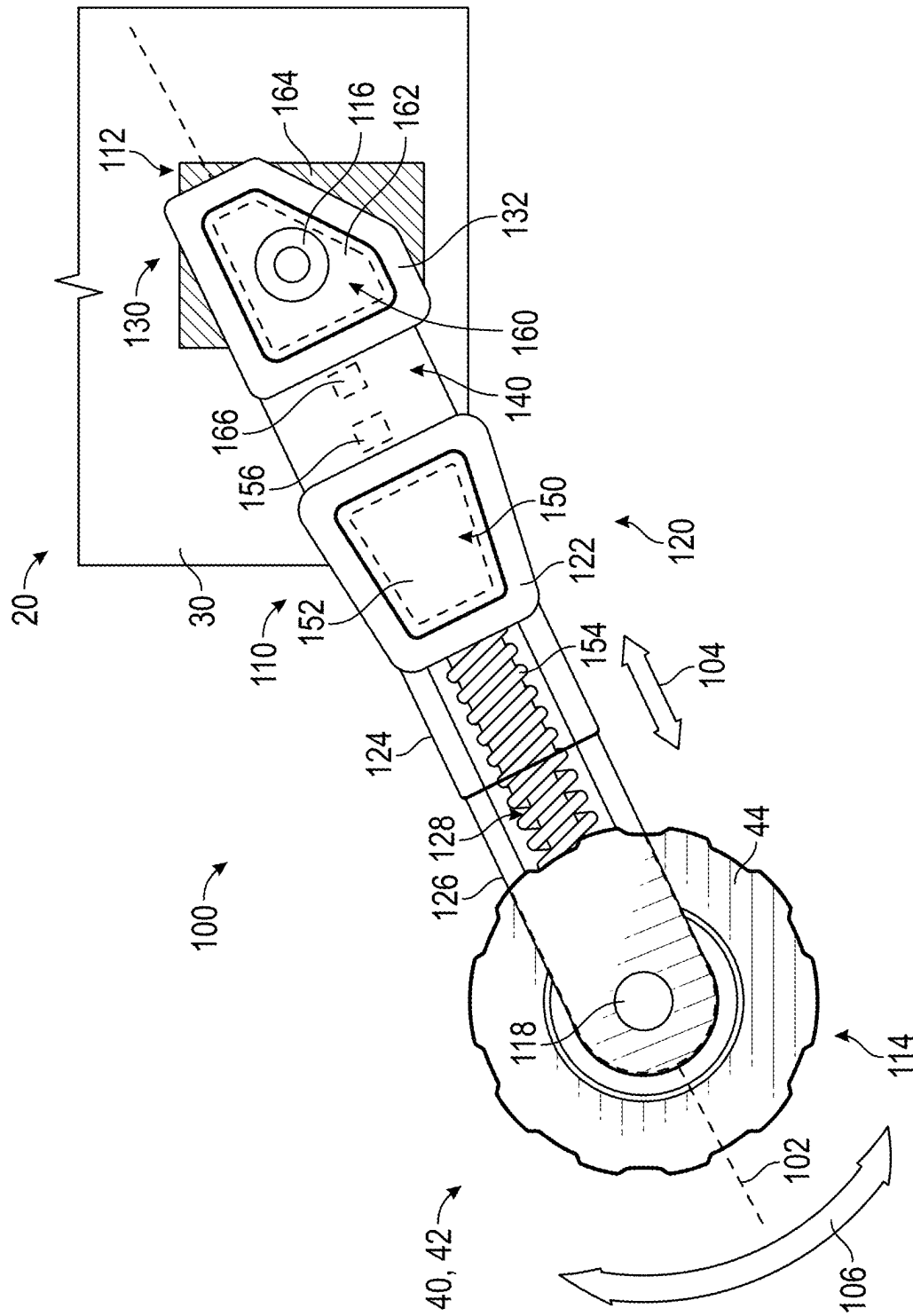


FIG. 3

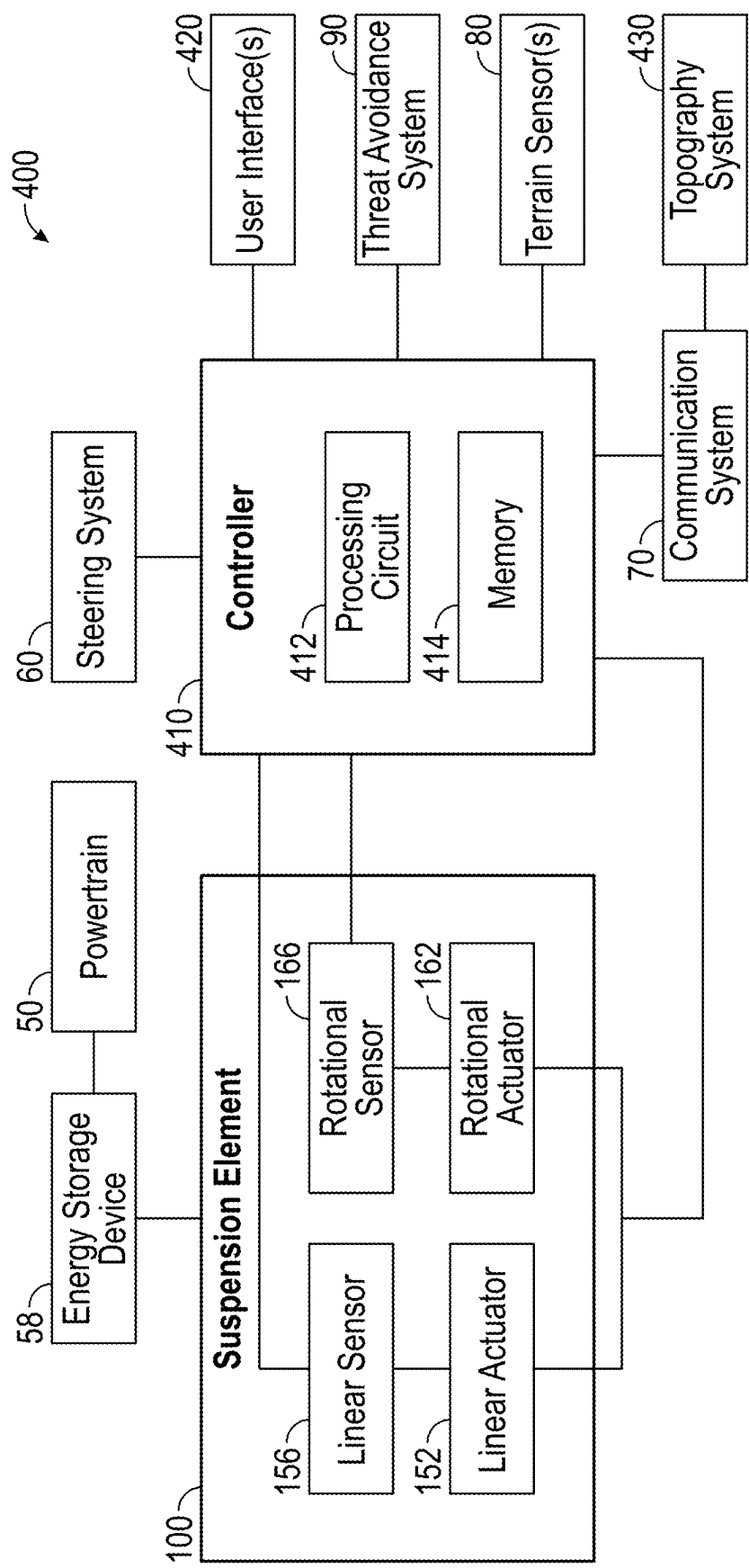


FIG. 4

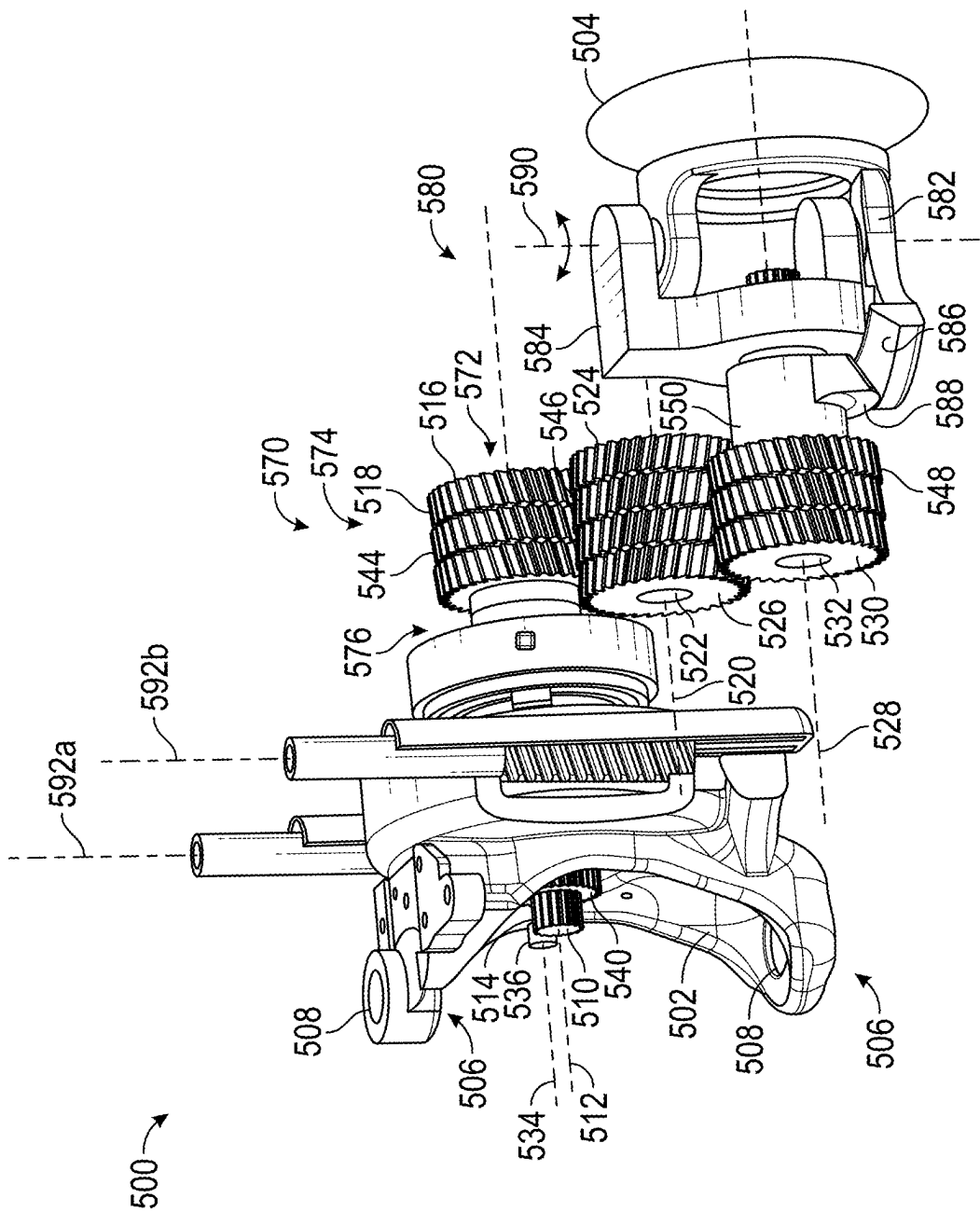


FIG. 5

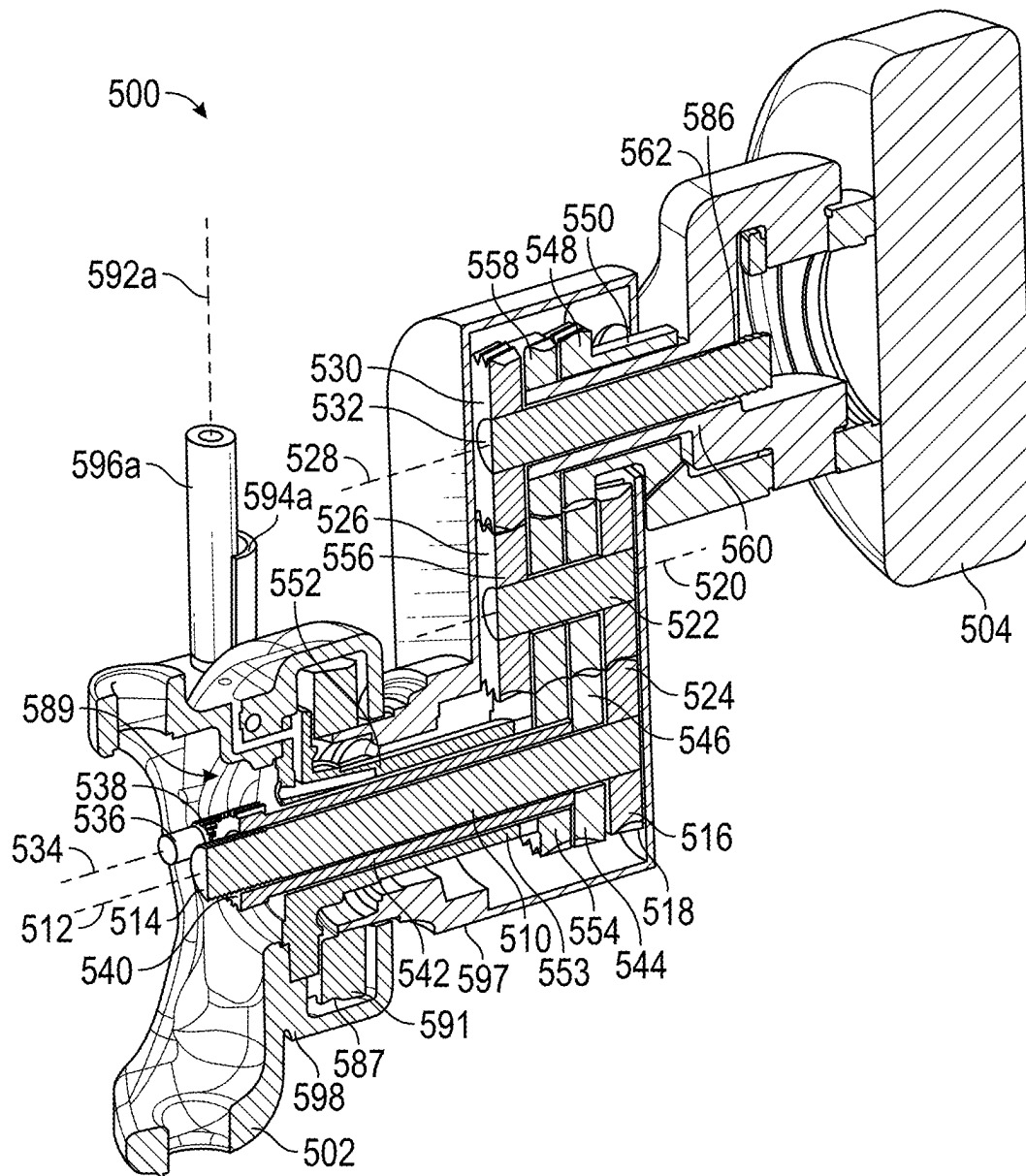


FIG. 6

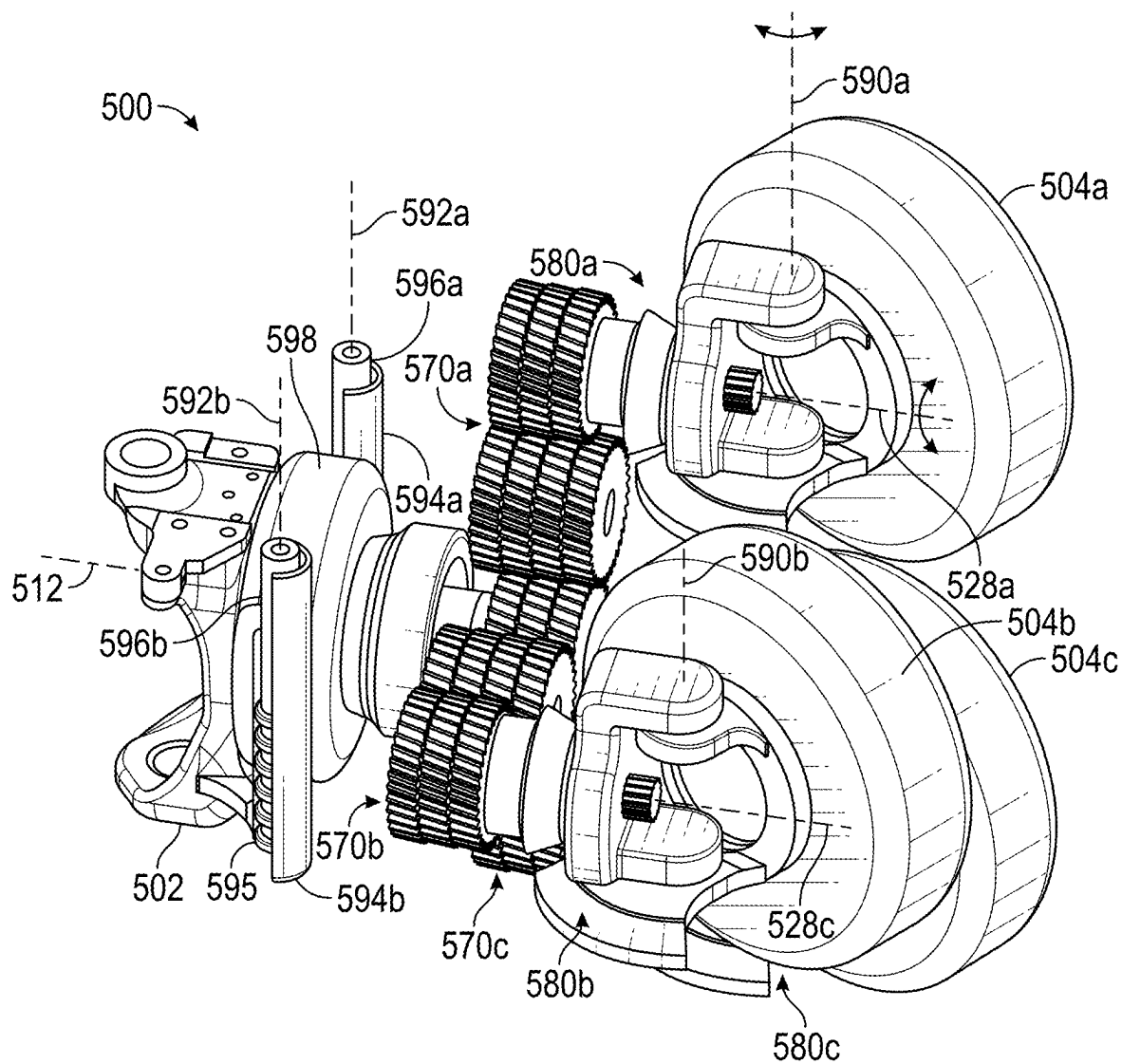


FIG. 7

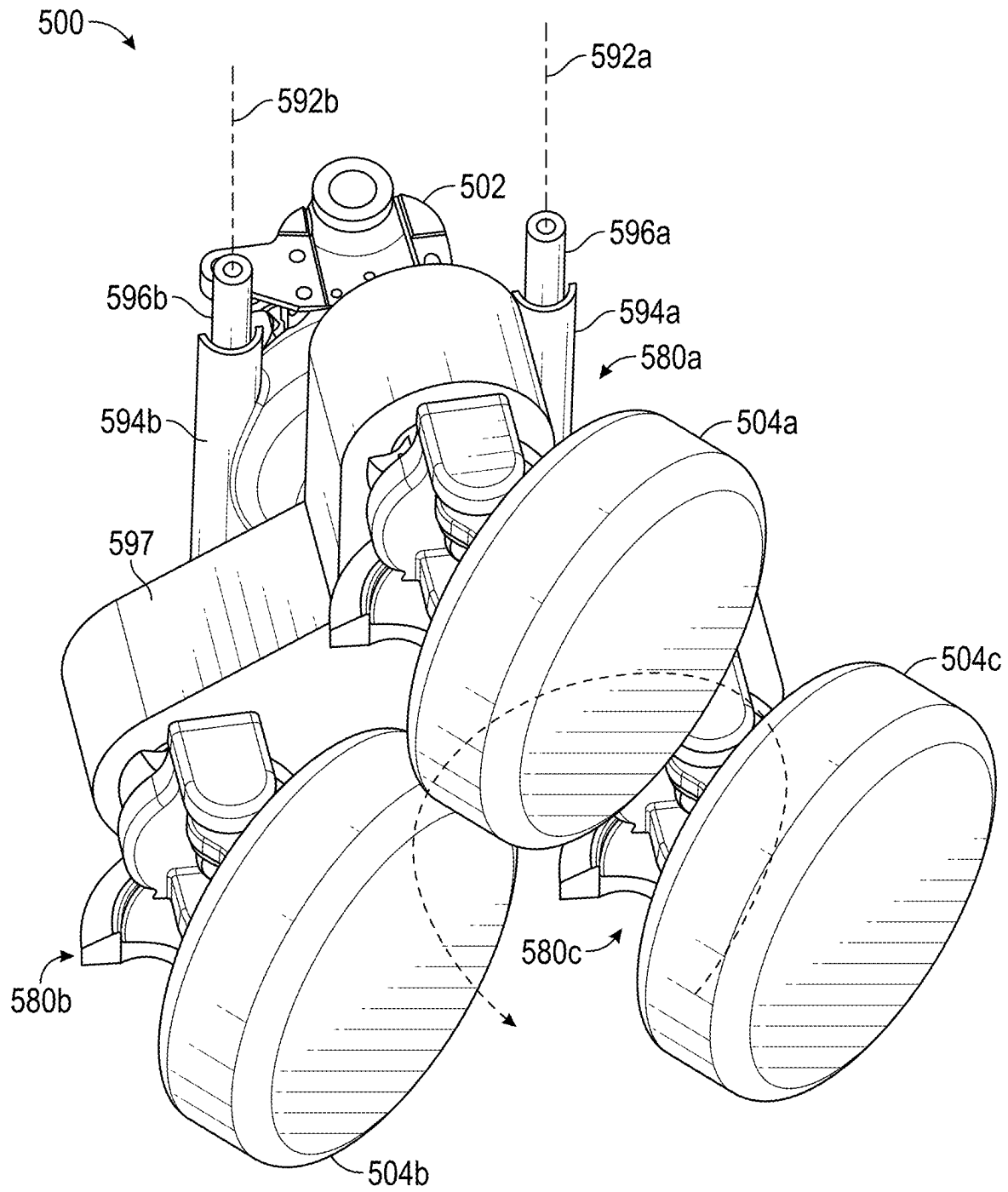


FIG. 8

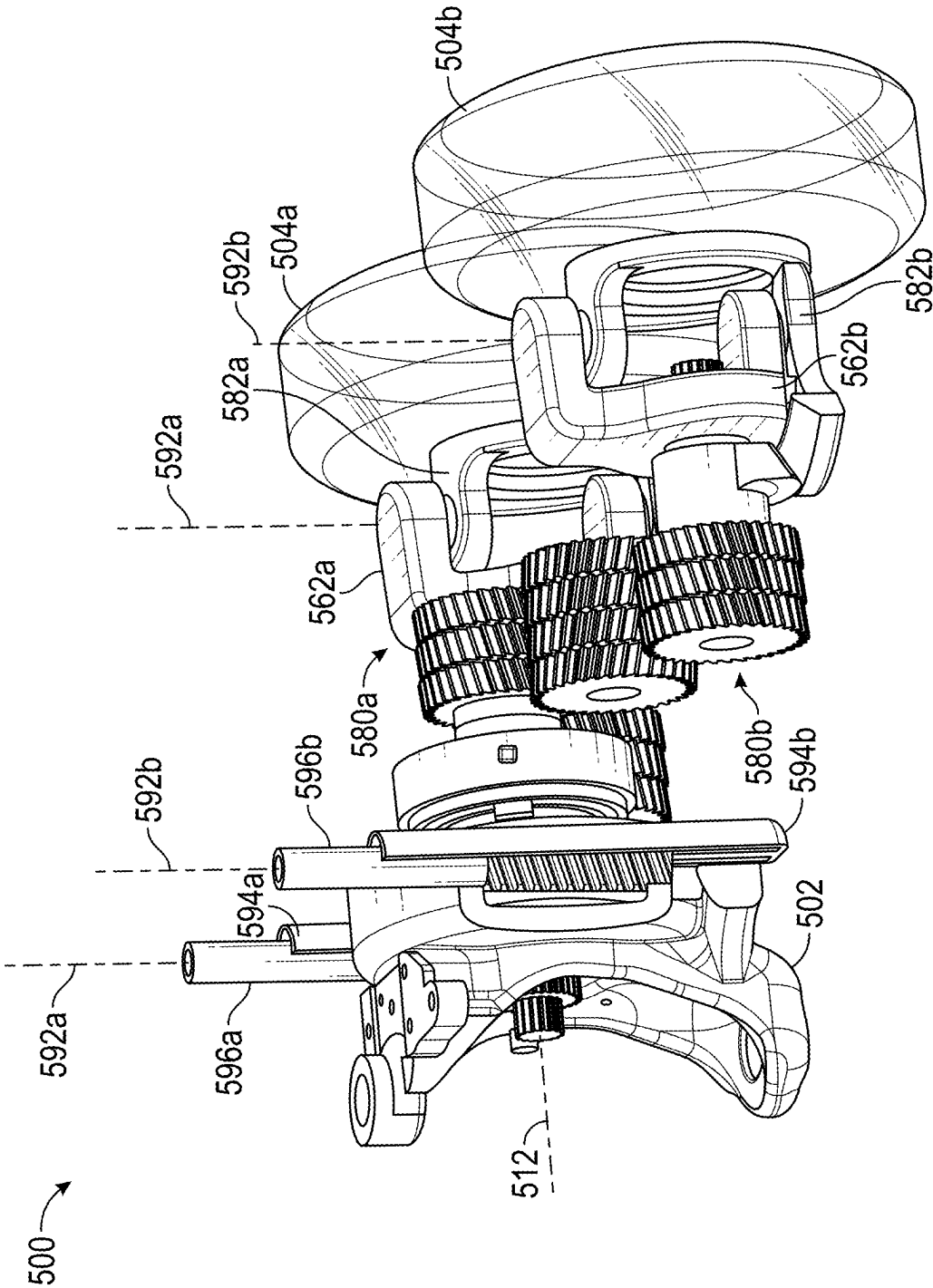


FIG. 9

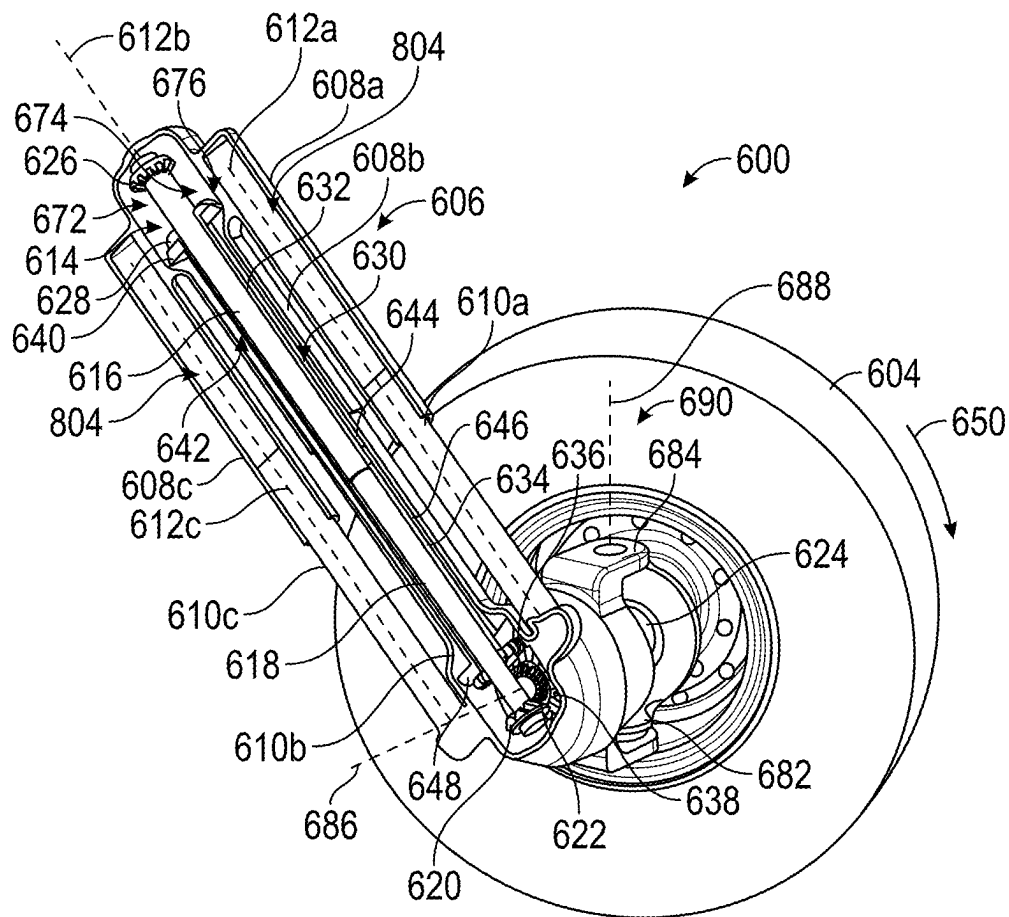


FIG. 10

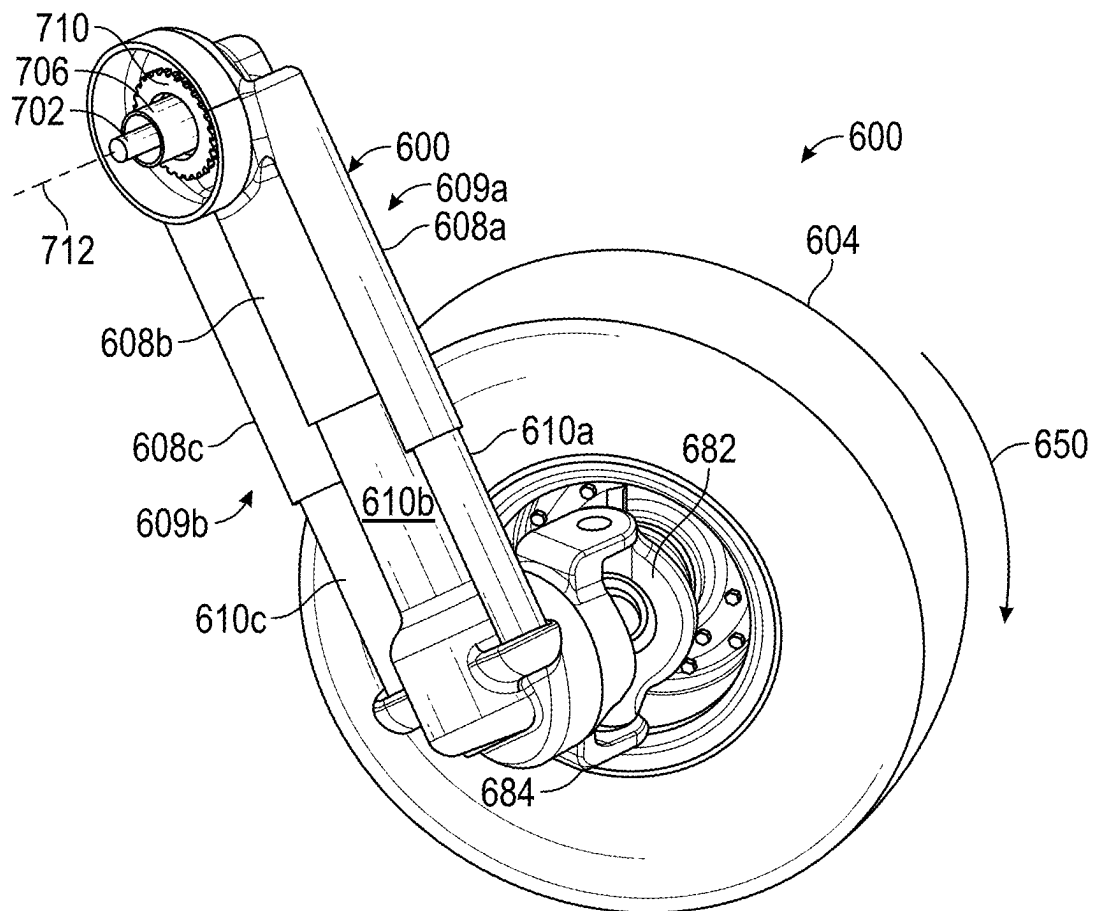


FIG. 11

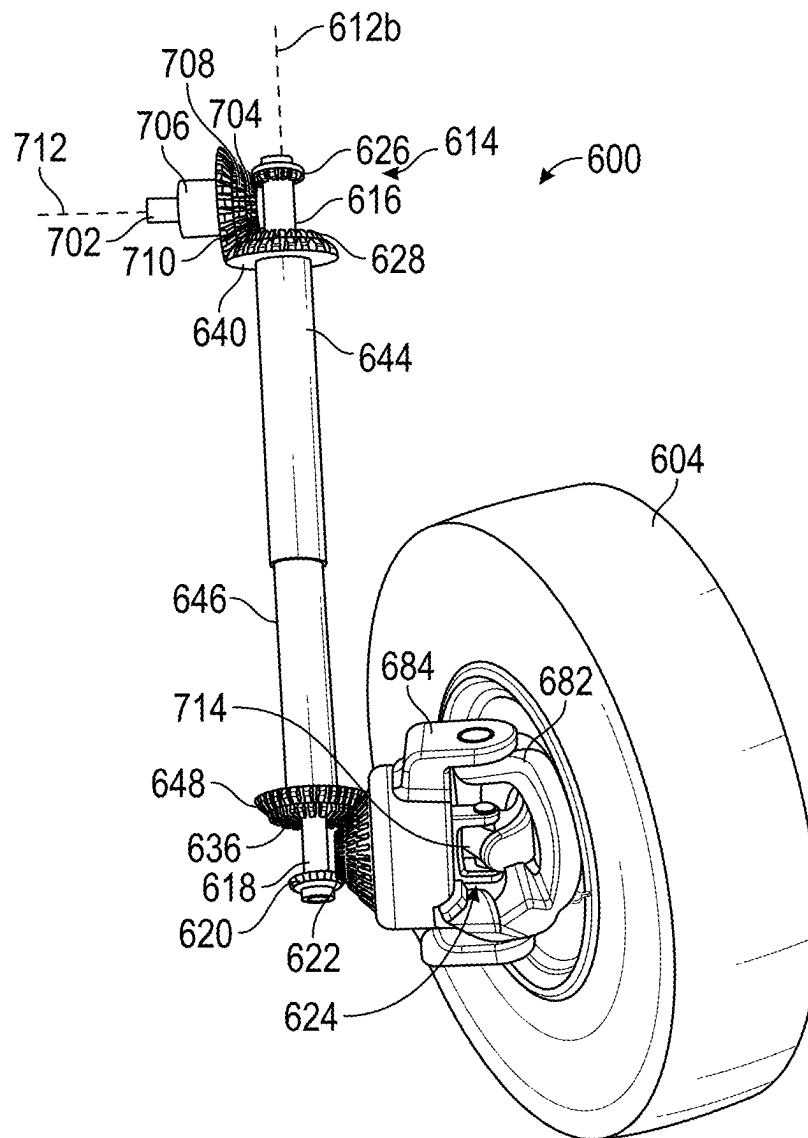


FIG. 12

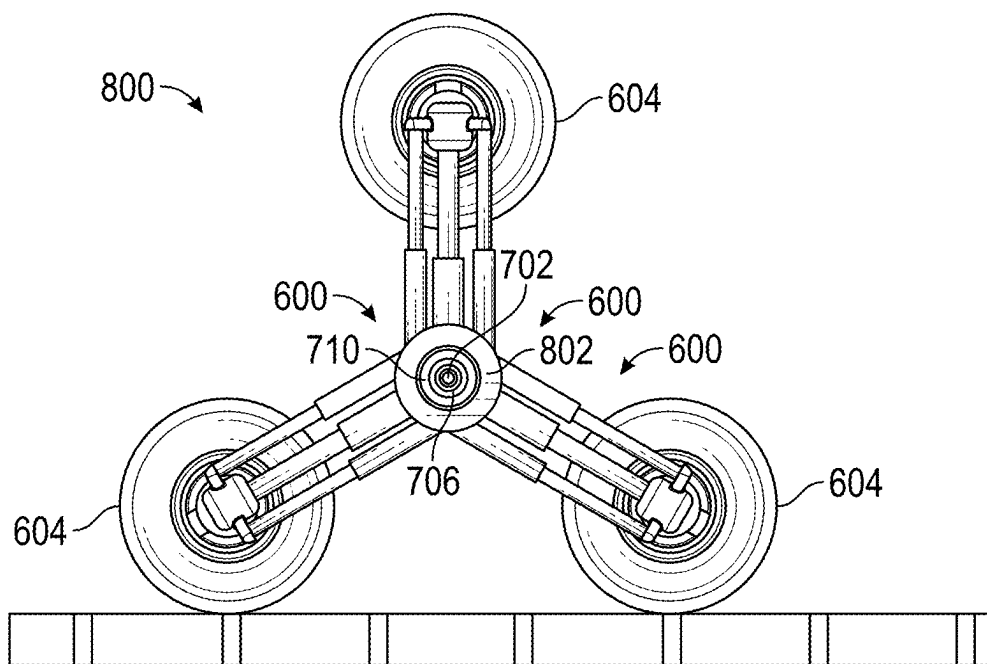


FIG. 13

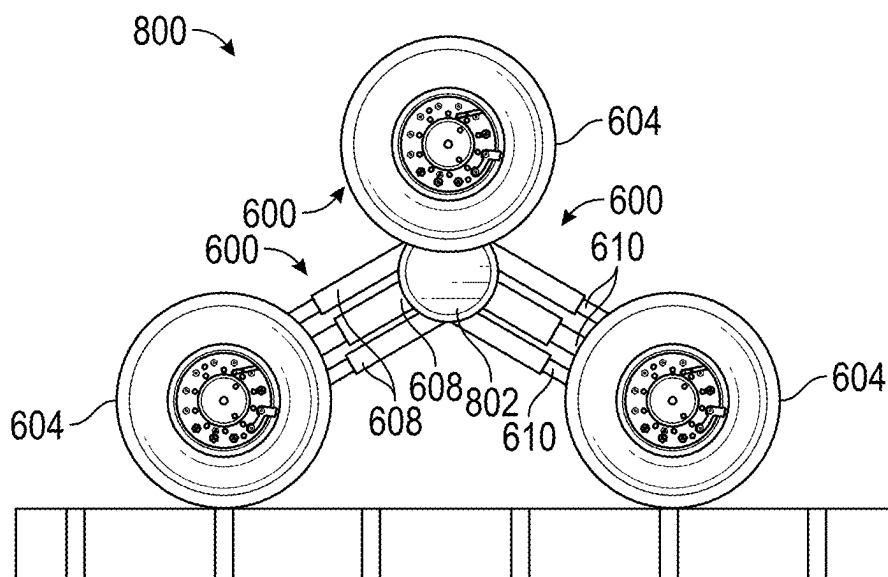


FIG. 14

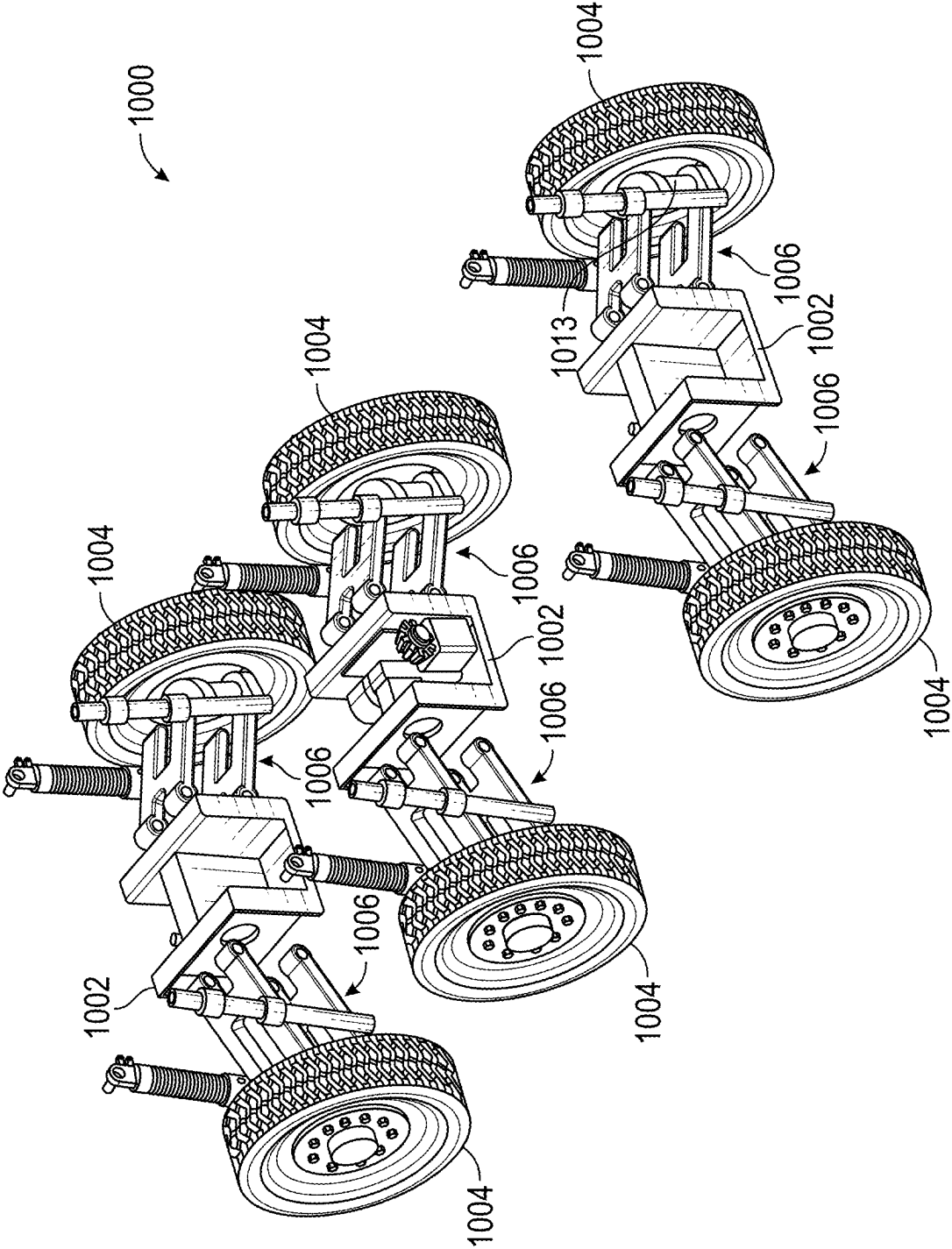


FIG. 15

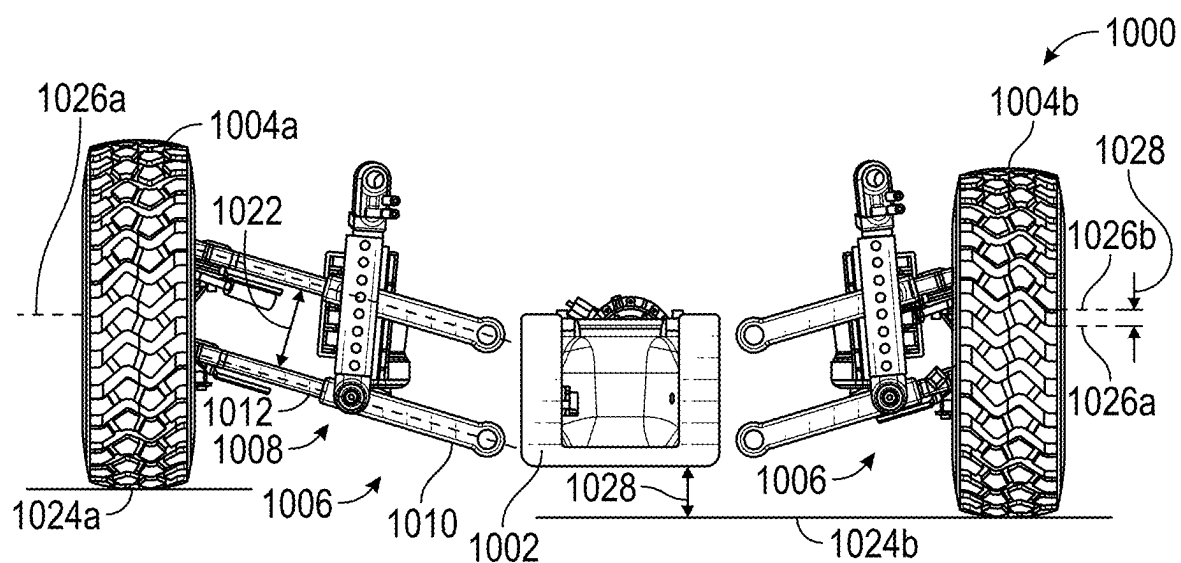


FIG. 16

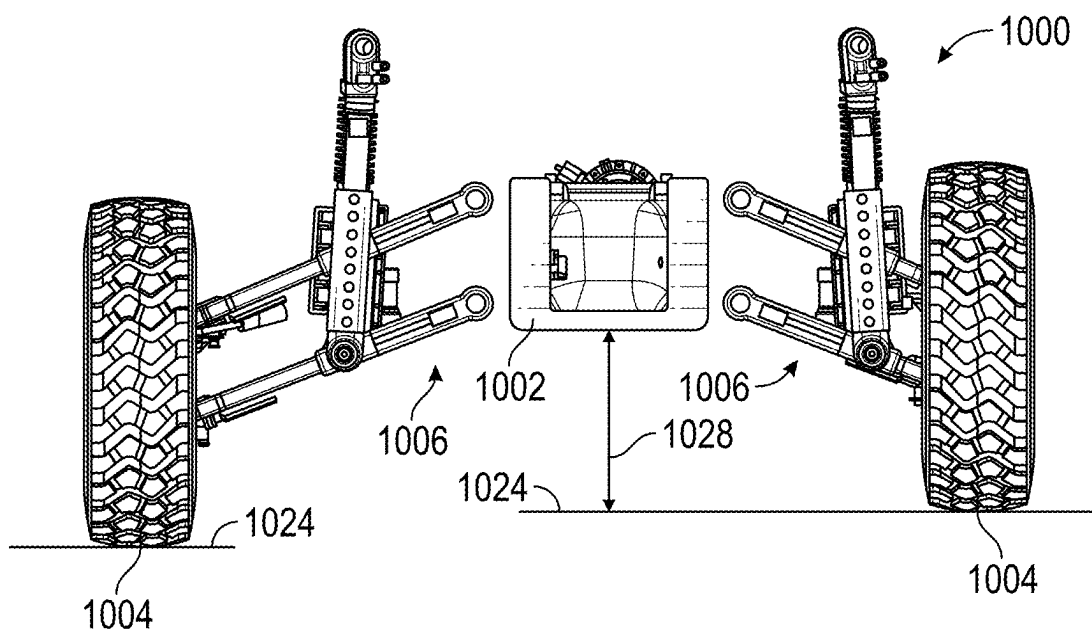


FIG. 17

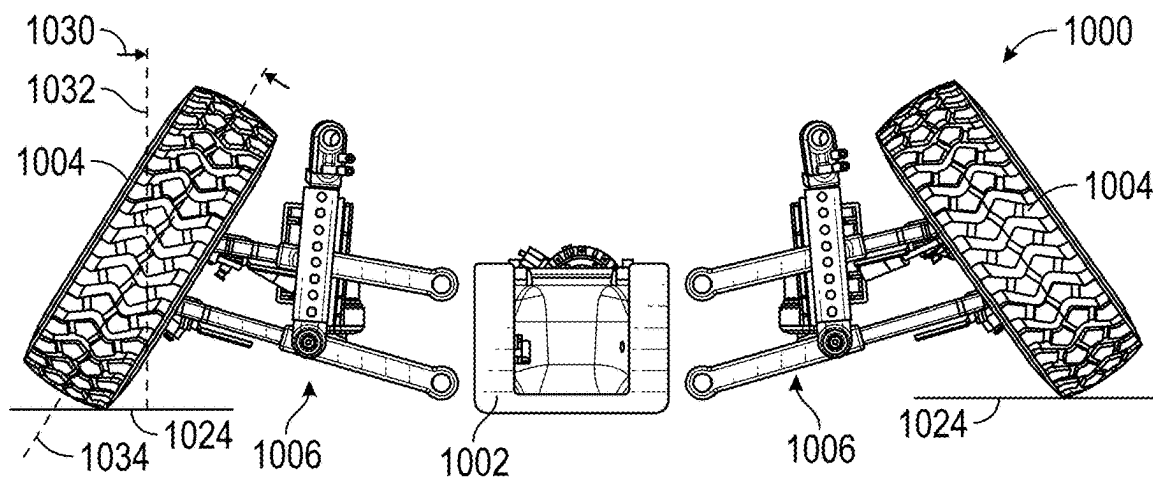


FIG. 18

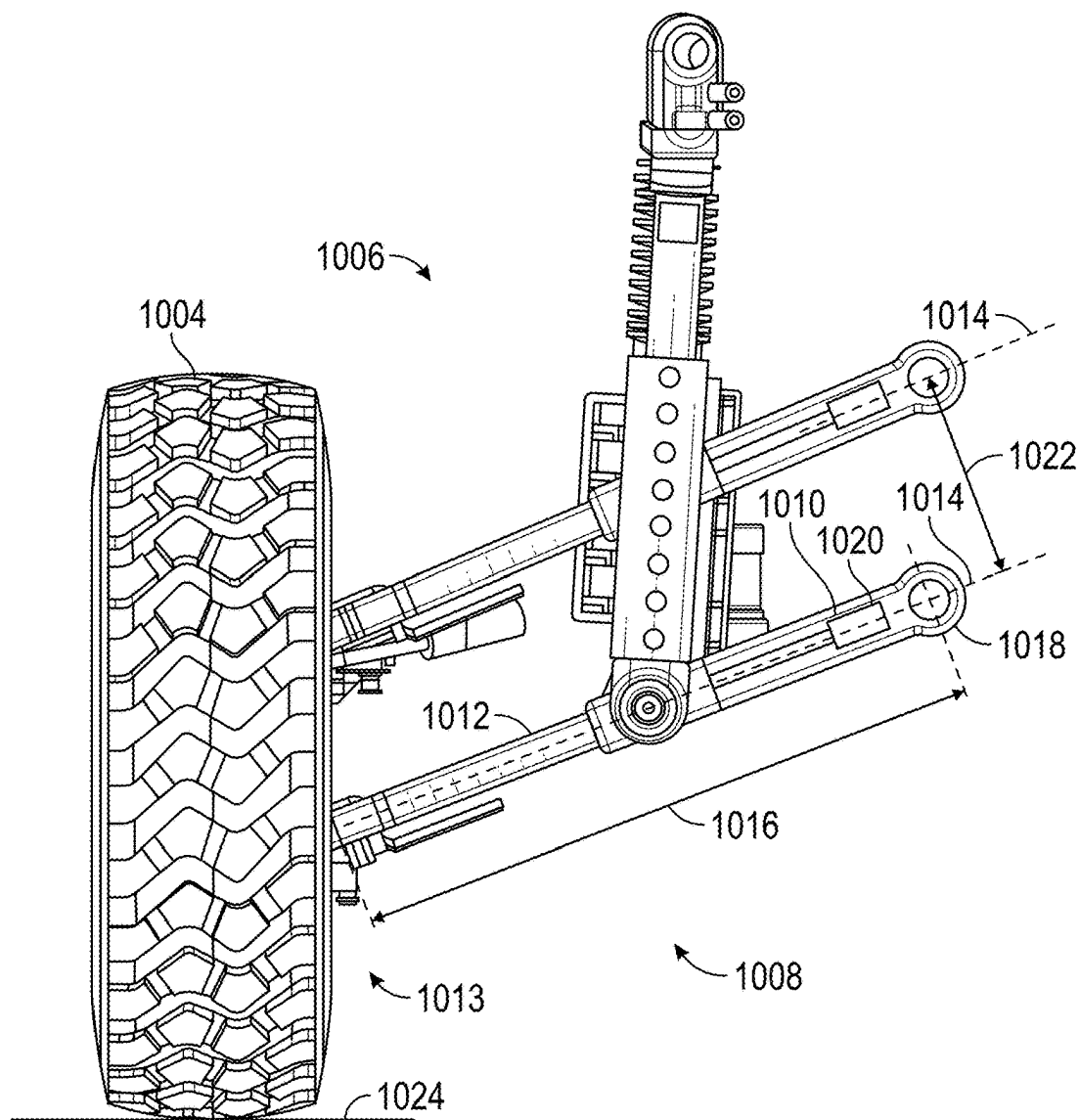


FIG. 19

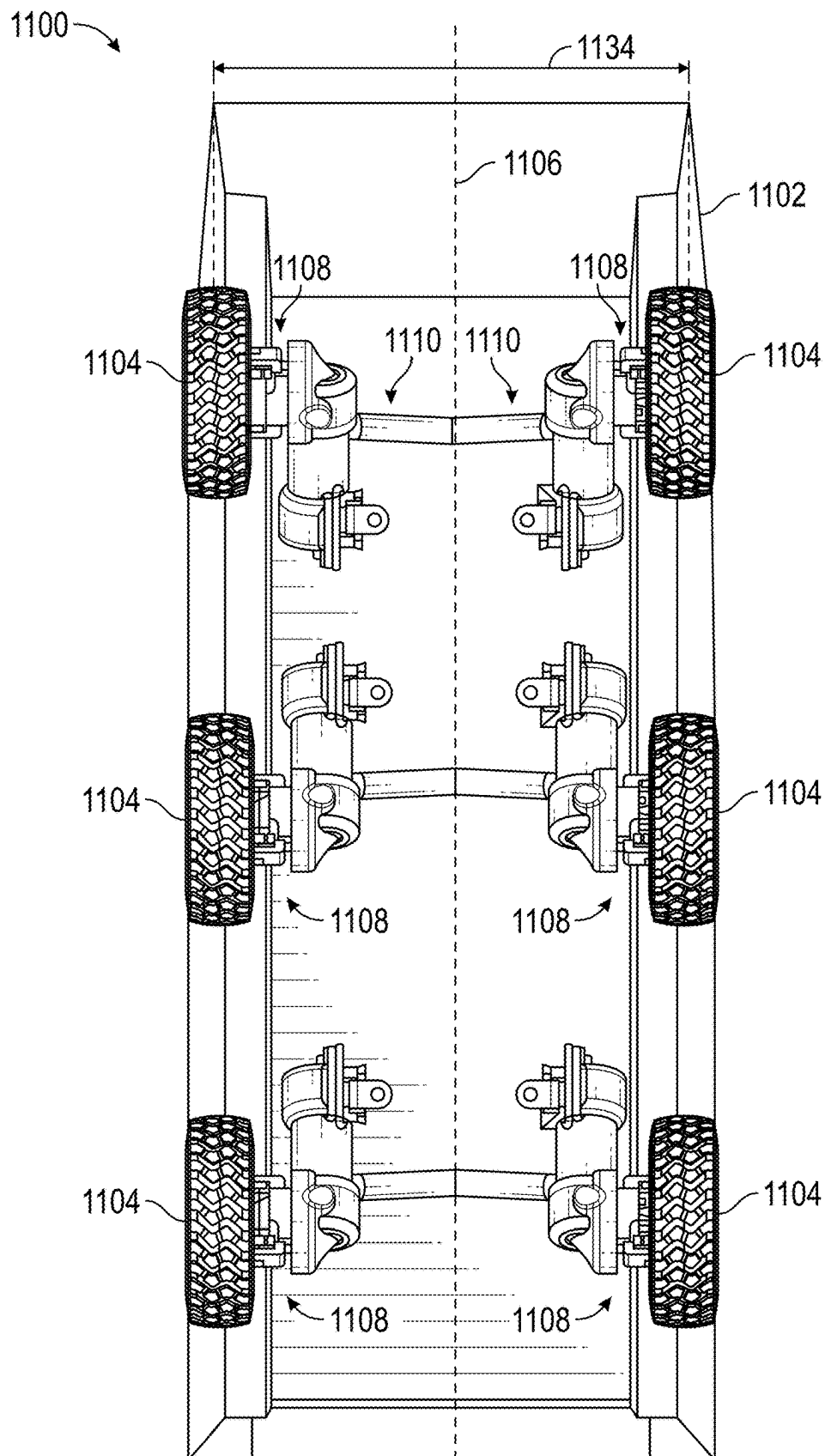


FIG. 20

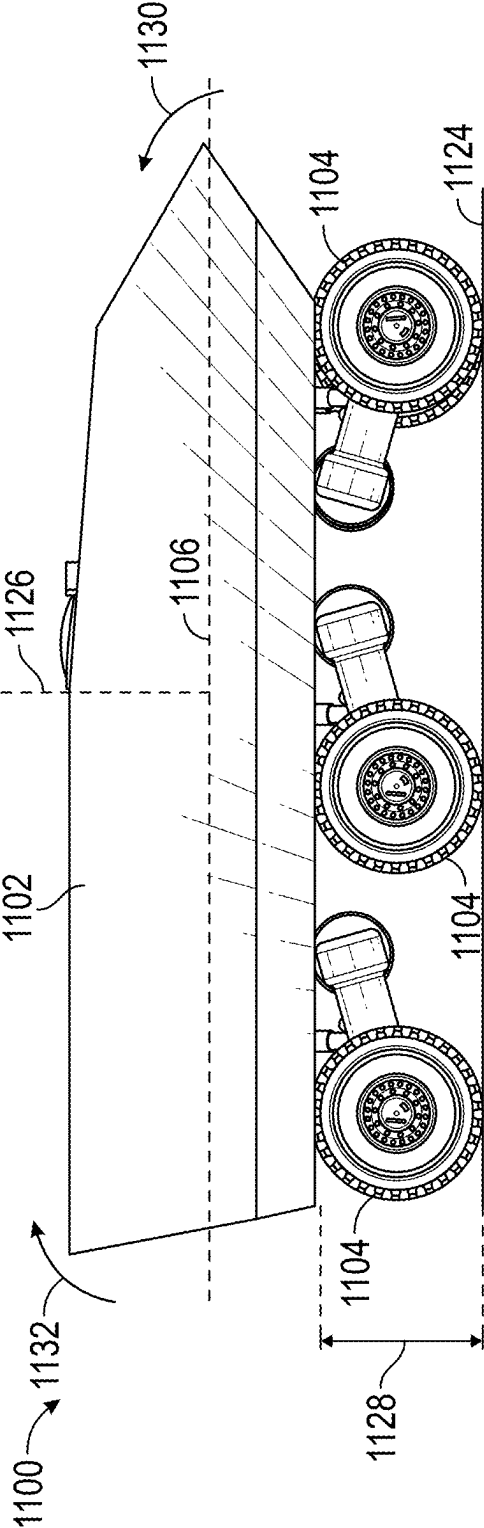


FIG. 21

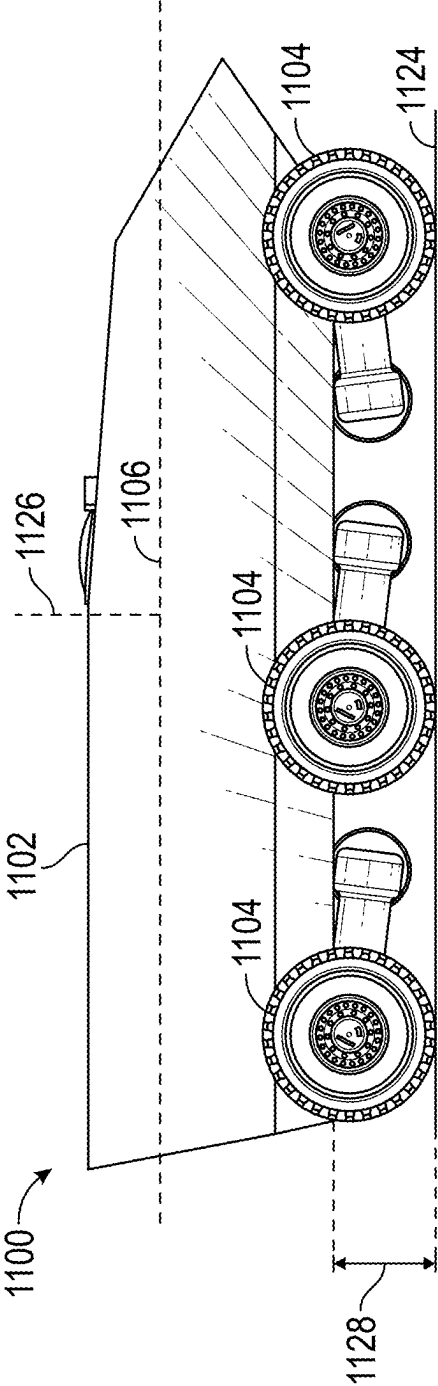


FIG. 22

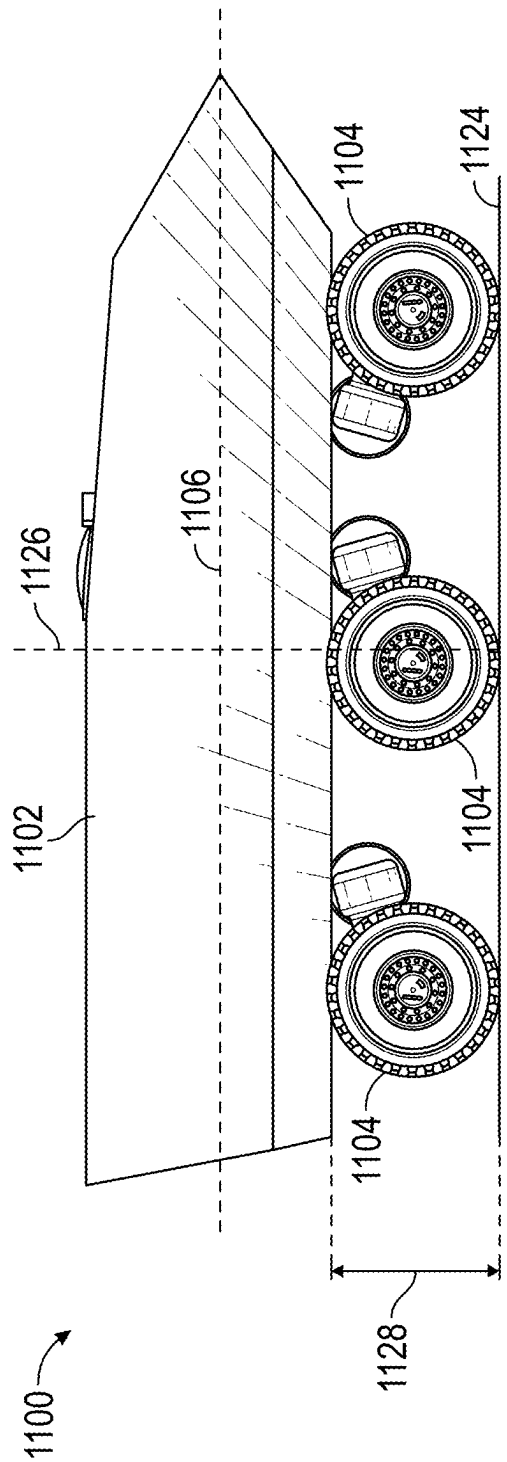


FIG. 23

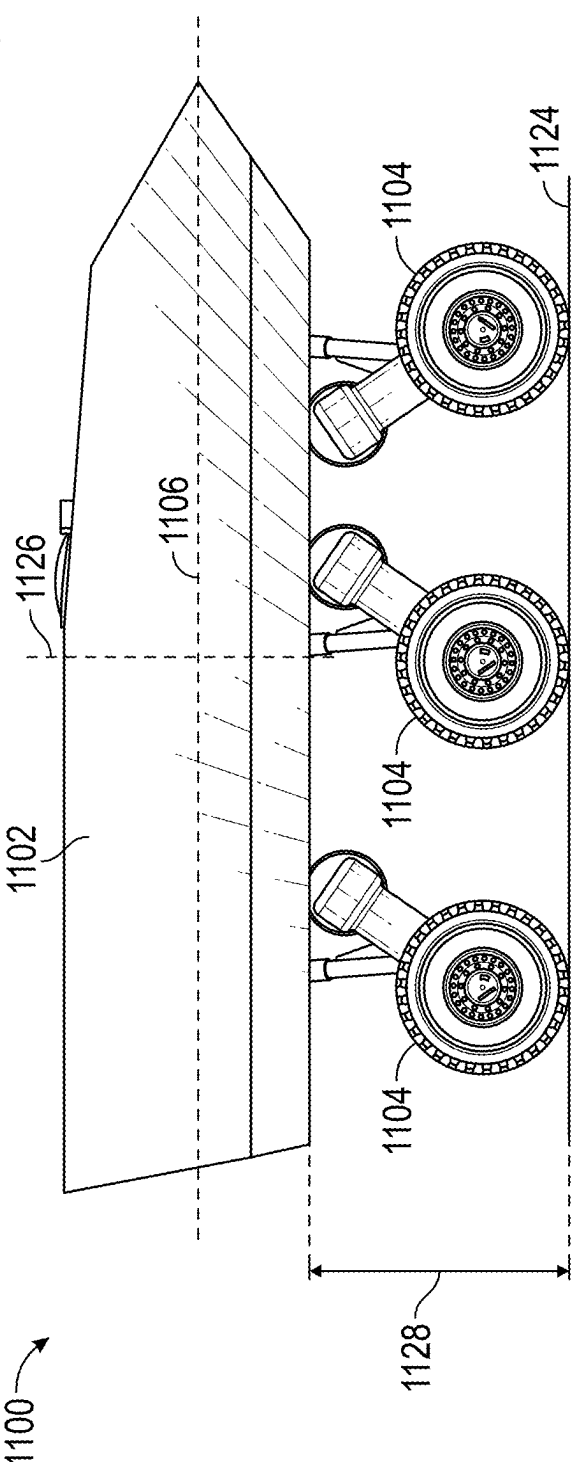


FIG. 24

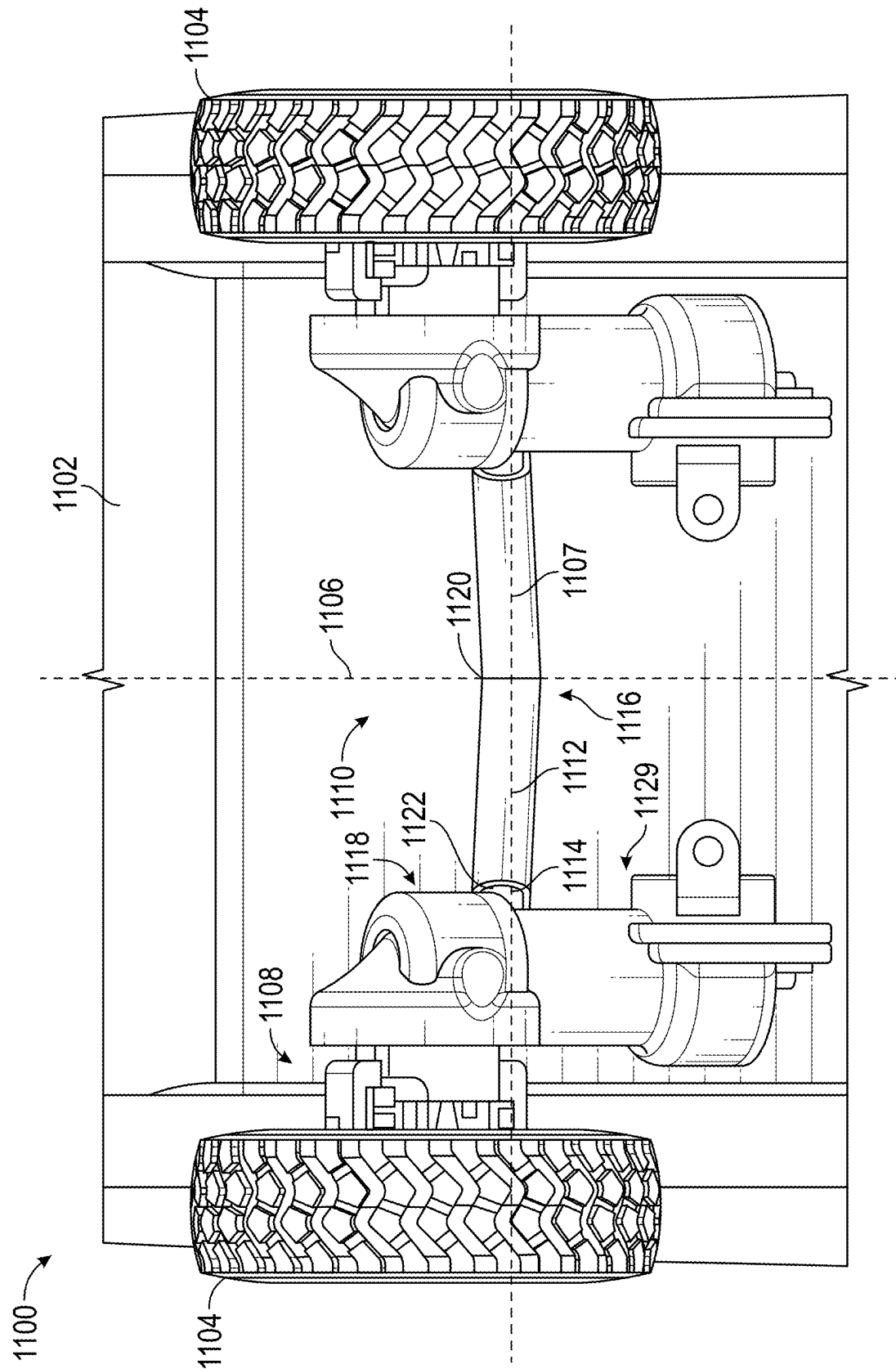


FIG. 25

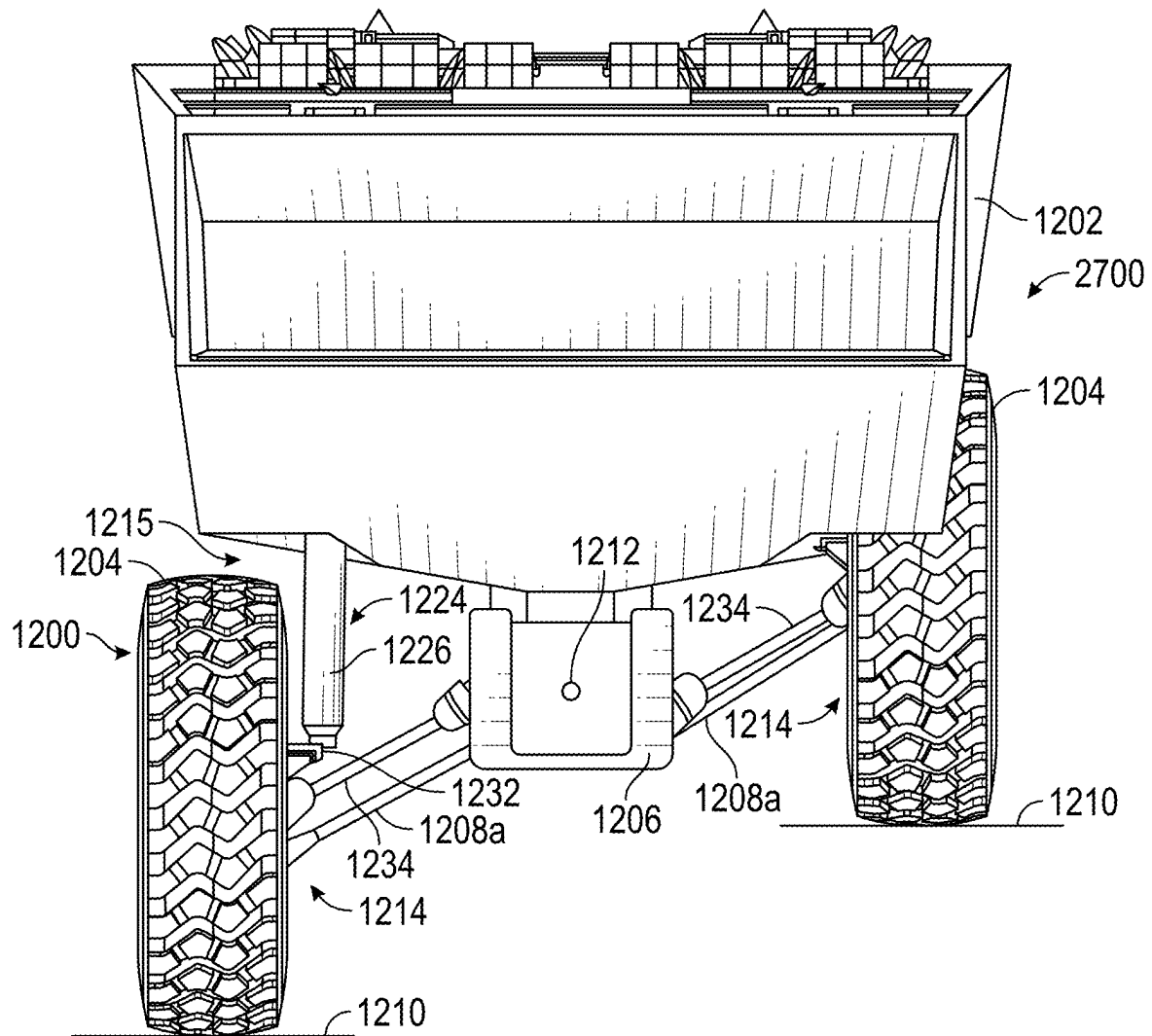


FIG. 26

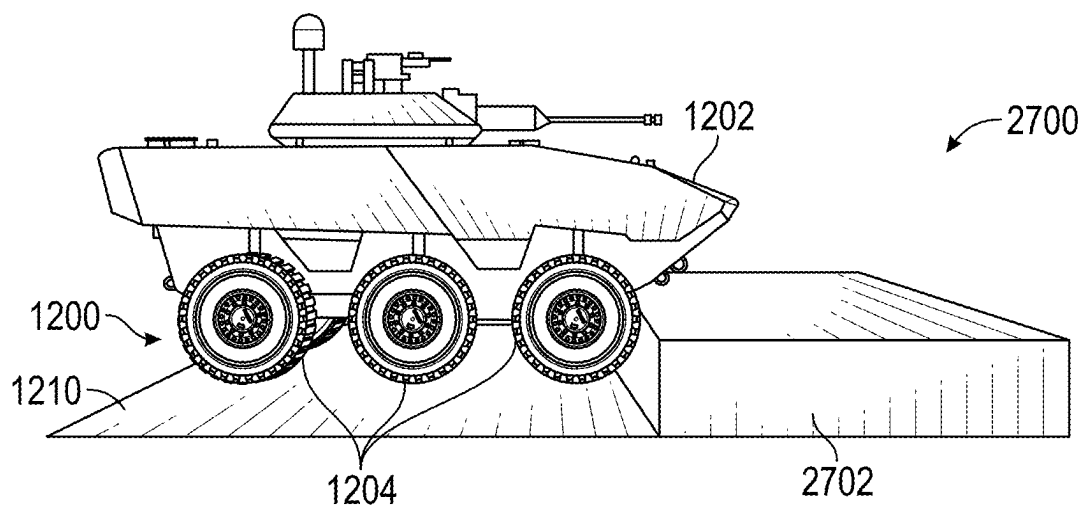


FIG. 27

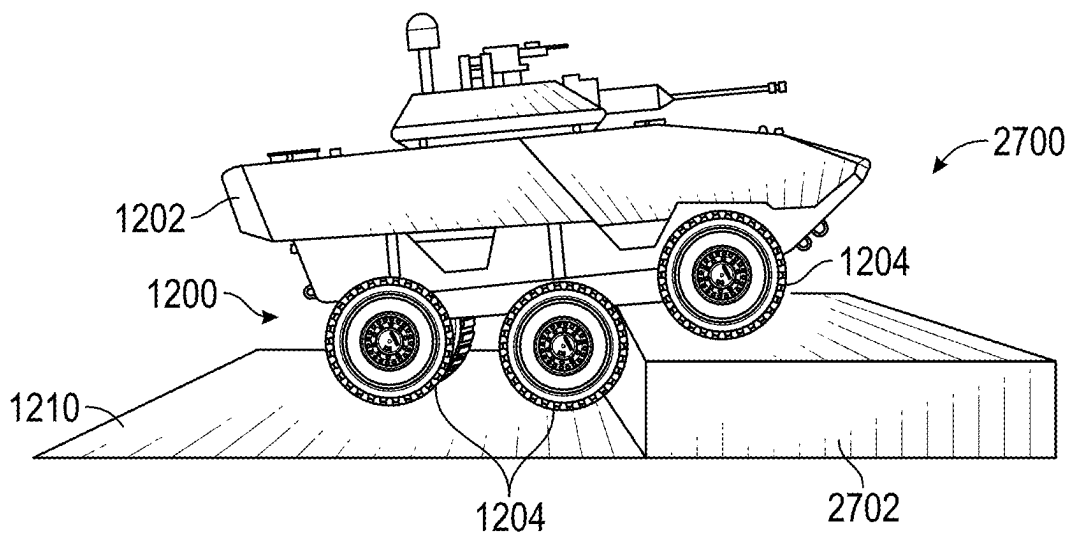


FIG. 28

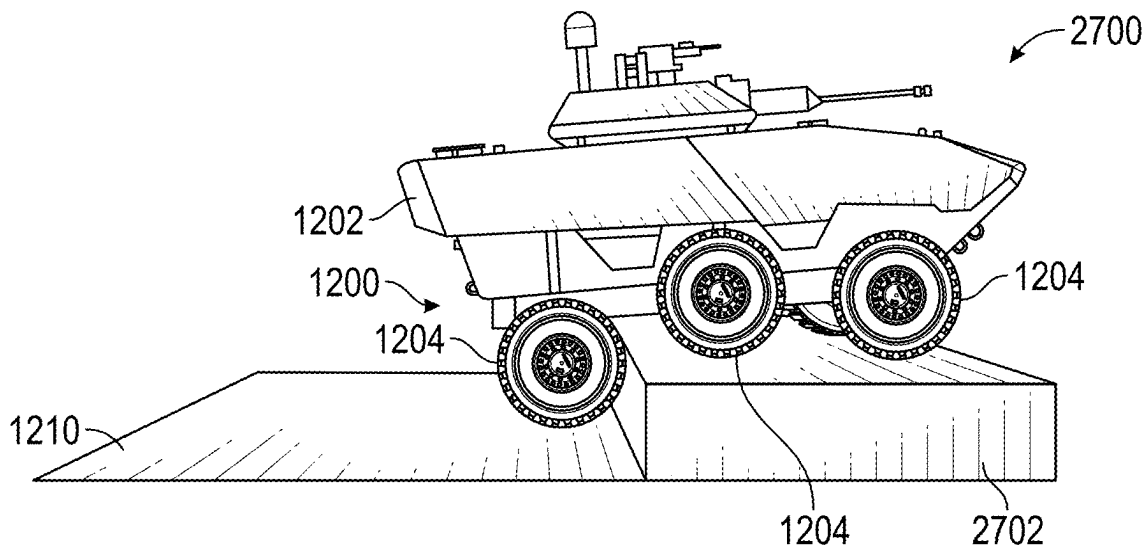


FIG. 29

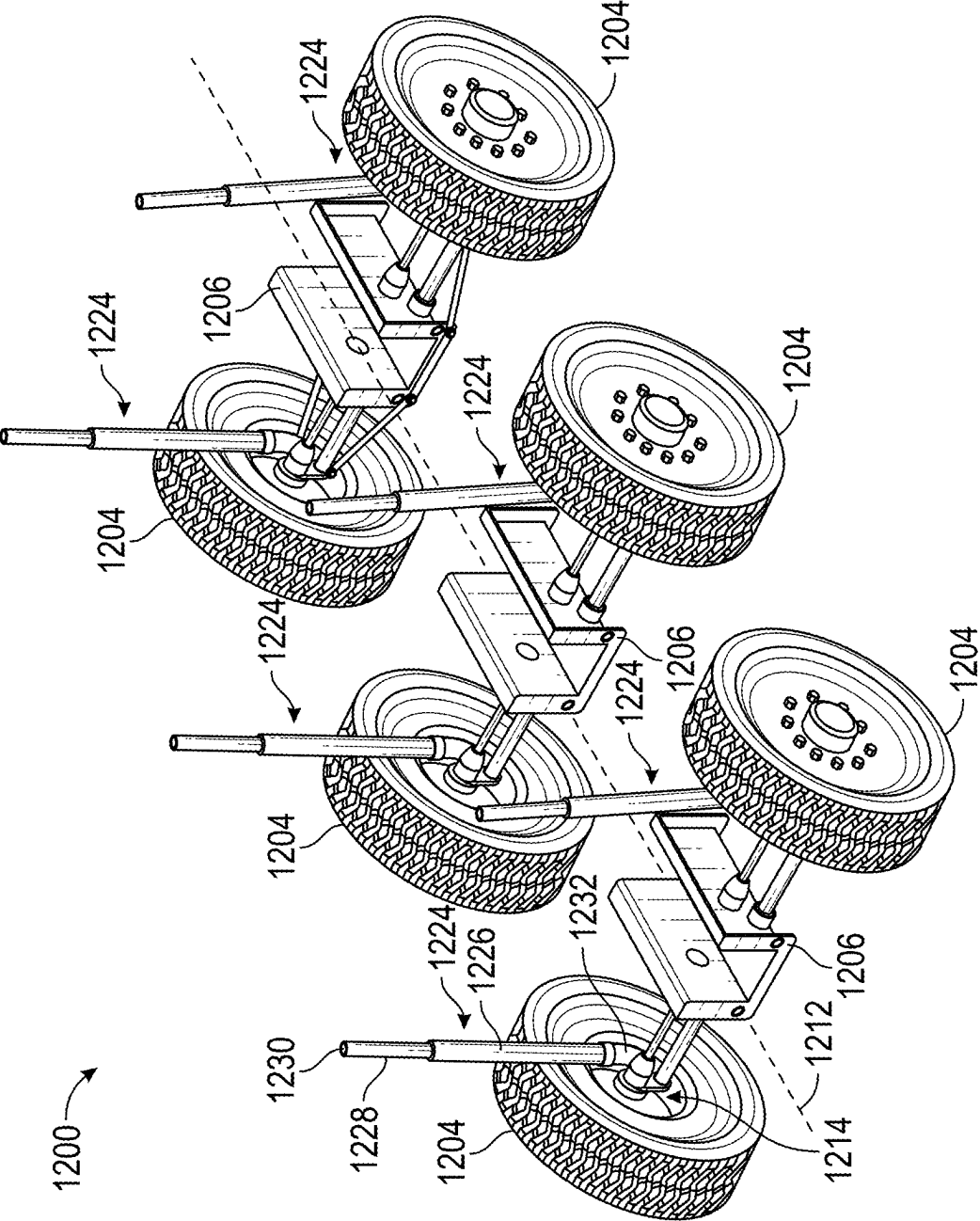


FIG. 30

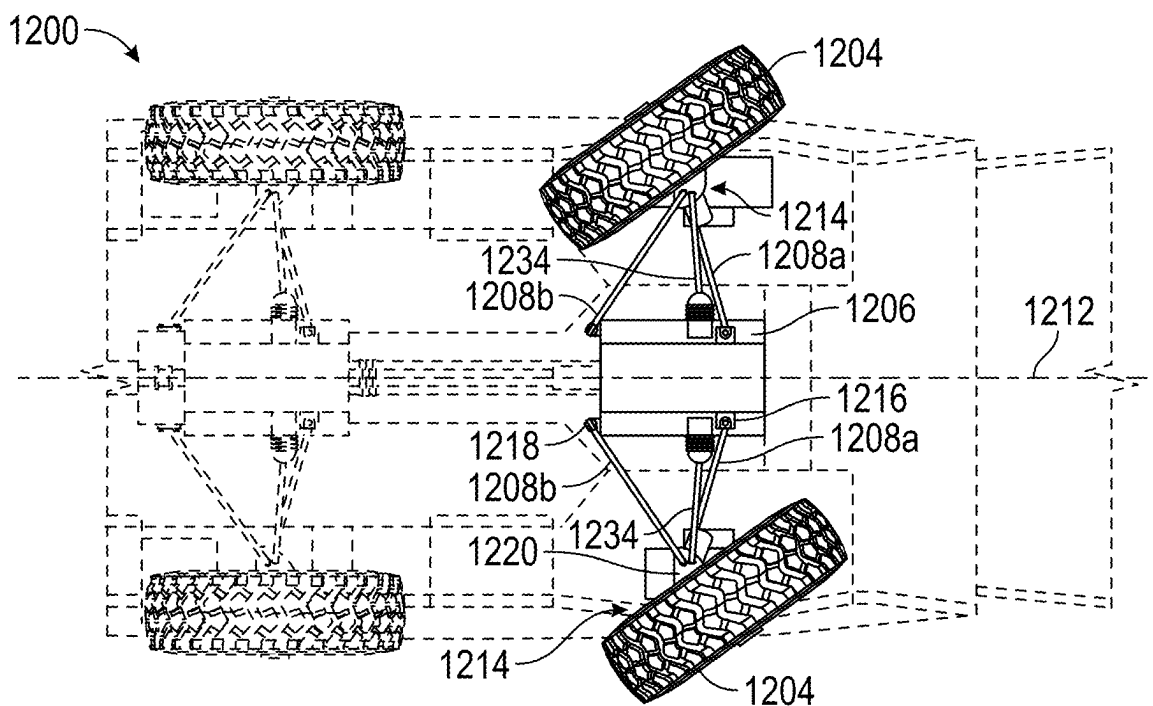


FIG. 31

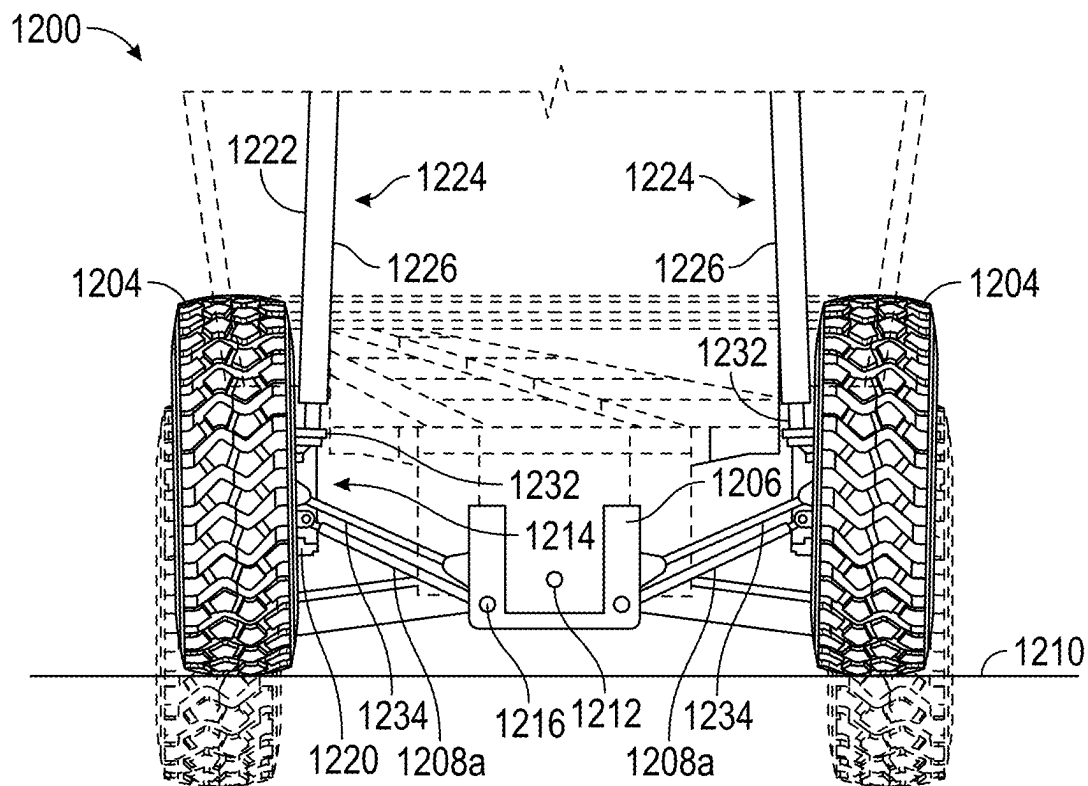


FIG. 32

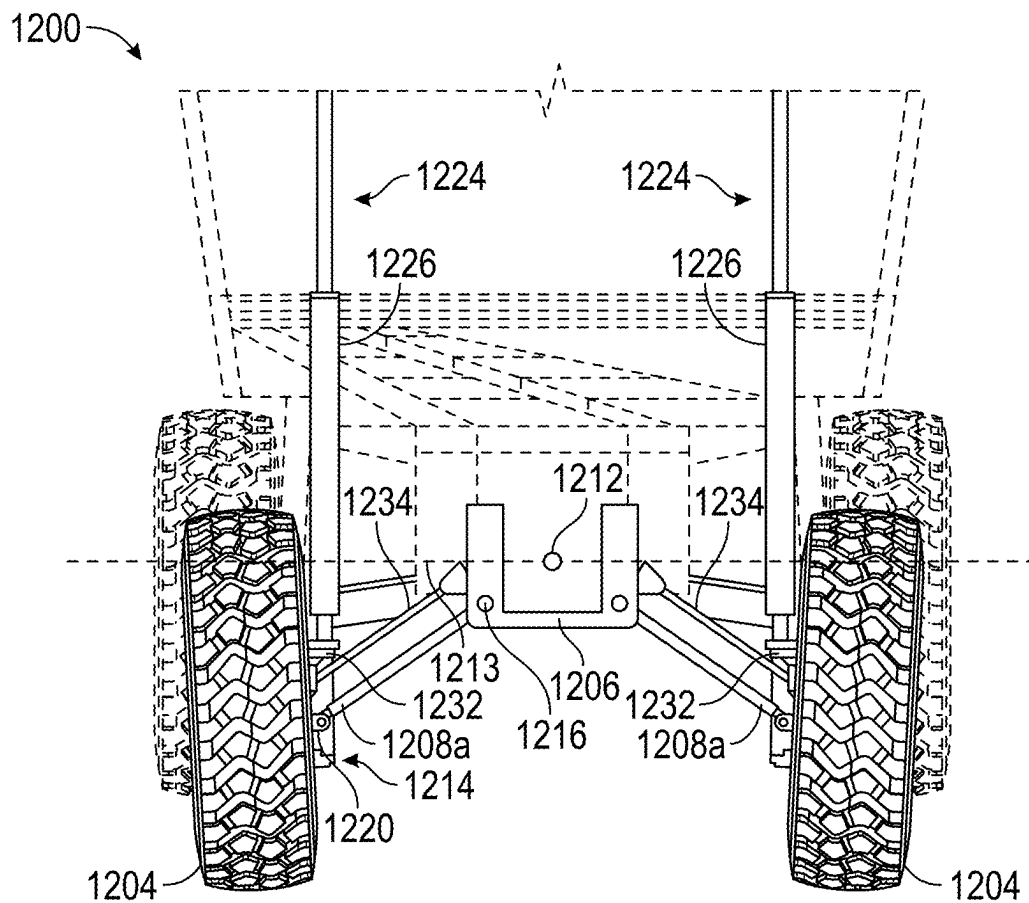


FIG. 33

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SYSTEMS AND METHODS FOR VEHICLE SUSPENSIONS

CROSS-REFERENCE TO RELATED PATENT APPLICATION

This application is a continuation of U.S. application Ser. No. 18/209,875, filed Jun. 14, 2023, which is a continuation of U.S. application Ser. No. 17/097,605, filed Nov. 13, 2020, which is a continuation of U.S. patent application Ser. No. 17/085,748, filed Oct. 30, 2020, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/927,852, filed Oct. 30, 2019, all of which are incorporated herein by reference in their entireties.

BACKGROUND

Suspension systems traditionally couple a body of a vehicle to one or more axles. Such suspension systems may include solid axle suspension systems and independent suspension systems, among others. Independent suspension systems facilitate independent wheel movement as the vehicle encounters one or more obstacles (e.g., uneven terrain, potholes, curbs, etc.). The independent suspension system reduces the forces experienced by passengers as the vehicle encounters the obstacles. Independent suspension systems may include one or more arms (e.g., A-arms, swing arms, etc.) that are coupled to a hub. A wheel and tire assembly may be attached to the hub. Various suspension components are coupled to the arms and the body of the vehicle.

SUMMARY

One embodiment of the present disclosure is a suspension system for a vehicle. The suspension system includes a wheel hub, a tractive element, a housing, a first power transmission train, and a second power transmission train. The tractive element is rotatably coupled with the wheel hub and configured to be driven for transportation and rotated for steering. The first power transmission train is positioned at least partially within the housing and is configured to drive the tractive element for transportation. The first power transmission train includes a first telescoping drive shaft. The second power transmission train is positioned at least partially within the housing and is configured to drive the tractive element to rotate for steering. The second power transmission train includes a second telescoping drive shaft. The second telescoping drive shaft is positioned within the first telescoping drive shaft, and both the first telescoping drive shaft and the second telescoping drive shaft are positioned within the housing.

Another embodiment of the present disclosure is a suspension system for a vehicle. The suspension system includes multiple suspension sub-systems. Each of the multiple suspension sub-systems are angularly offset from each other. Each of the multiple suspension sub-systems include a wheel hub, a tractive element, a housing, and multiple power transmission trains. The tractive element is rotatably coupled with the wheel hub and is configured to be driven for transportation and rotated for steering. The multiple power transmission trains are positioned at least partially within the housing. Each of the power transmission trains are configured to independently drive the tractive element for transportation, to drive the tractive element to rotate for steering, or to drive the wheel hub to rotate.

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Another embodiment of the present disclosure is a vehicle including a chassis and a suspension system coupled with the chassis. The suspension system includes a wheel hub, a tractive element, a housing, a first power transmission train, and a second power transmission train. The tractive element is rotatably coupled with the wheel hub and configured to be driven for transportation and rotated for steering. The first power transmission train is positioned at least partially within the housing and is configured to drive the tractive element for transportation. The first power transmission train includes a first telescoping drive shaft. The second power transmission train is positioned at least partially within the housing configured to drive the tractive element to rotate for steering. The second power transmission train includes a second telescoping drive shaft. The second telescoping drive shaft is positioned within the first telescoping drive shaft, and both the first telescoping drive shaft and the second telescoping drive shaft are positioned within the housing.

This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, in which:

FIG. 1 is a perspective view of a vehicle, according to an exemplary embodiment;

FIG. 2 is a schematic side view of the vehicle of FIG. 1, according to an exemplary embodiment;

FIG. 3 is a schematic side view of a suspension element of the vehicle of FIG. 1, according to an exemplary embodiment;

FIG. 4 is a schematic block diagram of a controller configured to operate and/or control various components of a vehicle, according to an exemplary embodiment;

FIG. 5 is a perspective view of a suspension system, according to an exemplary embodiment;

FIG. 6 is a cross-sectional perspective view of the suspension system of FIG. 5, according to an exemplary embodiment;

FIG. 7 is a perspective view of the suspension system of FIG. 5 configured for three tractive elements, according to an exemplary embodiment;

FIG. 8 is a perspective view of the suspension system of FIG. 7, according to an exemplary embodiment;

FIG. 9 is a perspective view of the suspension system of FIG. 5, configured for two tractive elements, according to an exemplary embodiment;

FIG. 10 is a perspective view of a cross section of a suspension system that uses concentric shafts to steer, rotate a wheel hub, and drive a tractive element, according to an exemplary embodiment;

FIG. 11 is a perspective view of the suspension system of FIG. 10, according to an exemplary embodiment;

FIG. 12 is a perspective view of the suspension system of FIG. 10, according to an exemplary embodiment;

FIG. 13 is a side view of a suspension system that uses three of the suspension systems of FIG. 10, according to an exemplary embodiment;

FIG. 14 is a side view of the suspension system of FIG. 13, according to an exemplary embodiment;

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FIG. 15 is a perspective view of a suspension system with an extendable length control arm, according to an exemplary embodiment;

FIG. 16 is a front view of the suspension system of FIG. 15, according to an exemplary embodiment;

FIG. 17 is a front view of the suspension system of FIG. 15, according to an exemplary embodiment;

FIG. 18 is a front view of the suspension system of FIG. 15, according to an exemplary embodiment;

FIG. 19 is a front view of a portion of the suspension system of FIG. 15, according to an exemplary embodiment;

FIG. 20 is a bottom view of an articulated arm suspension system, according to an exemplary embodiment;

FIG. 21 is a side view of the suspension system of FIG. 20 in a partially raised position, according to an exemplary embodiment;

FIG. 22 is a side view of the suspension system of FIG. 20 in a lowered position, according to an exemplary embodiment;

FIG. 23 is a side view of the suspension system of FIG. 20 in a partially lowered position, according to an exemplary embodiment;

FIG. 24 is a side view of the suspension system of FIG. 20 in a raised position, according to an exemplary embodiment;

FIG. 25 is a bottom view of a portion of the suspension system of FIG. 20, according to an exemplary embodiment;

FIG. 26 is a front view of a suspension system including ball jointed linkages, according to an exemplary embodiment;

FIG. 27 is a perspective view of a vehicle equipped with the suspension system of FIG. 26 approaching a step, according to an exemplary embodiment;

FIG. 28 is a perspective view of the vehicle of FIG. 27 crawling over the step, according to an exemplary embodiment;

FIG. 29 is a perspective view of the vehicle of FIG. 27 completing a crawl operation over the step, according to an exemplary embodiment;

FIG. 30 is a perspective view of the suspension system of FIG. 26, according to an exemplary embodiment;

FIG. 31 is a bottom view of a portion of the suspension system of FIG. 26, according to an exemplary embodiment;

FIG. 32 is a front view of a portion of the suspension system of FIG. 26, according to an exemplary embodiment; and

FIG. 33 is a front view of the suspension system of FIG. 26 with adjusted camber, according to an exemplary embodiment.

DETAILED DESCRIPTION

Before turning to the FIGURES, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the FIGURES. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

According to an exemplary embodiment, a vehicle includes a chassis and a body supported by a suspension system. Vehicle suspension systems may be passive, semi-active, or fully-active systems. Passive suspension systems may have conventional uncontrolled, passive springs and dampers configured to absorb and dissipate loads experienced by a tractive element (e.g., wheel assembly, etc.) to reduce (e.g., lessen, mitigate, etc.) impact loads (e.g., from

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bumps, potholes, etc.) transferred to the chassis, body, and/or passengers of the vehicle. Semi-active suspension systems may further reduce impact loads (e.g., relative to passive suspension systems, etc.) by controlling a damping force provided by the suspension system. According to an exemplary embodiment, the vehicle of the present disclosure has a fully-active suspension system configured to add energy with a controlled motoring force and remove energy with a controlled damping force. The suspension system has passive resilient members (e.g., springs, etc.) to support the static mass of the vehicle. Traditional passive, semi-active, and/or fully-active suspension systems are configured as single degree of freedom suspension systems (e.g., linear actuation, etc.).

According to an exemplary embodiment, the suspension system of the present disclosure is a fully-active and multiple degree of freedom suspension system configured to facilitate providing a near-constant force capability (e.g., the force transferred to the chassis, body, and/or passengers of the vehicle varies less than a threshold amount, is near-constant, etc.). The suspension system combines multiple degrees of freedom and active, independent suspension control to provide superior off-road performance and terrain accessibility of a vehicle (e.g., enabling access to terrain that current vehicles are not capable of negotiating, etc.). Equipped with the fully-active and multiple degree of freedom suspension system, the exemplary vehicle may no longer be limited to driving on roads or improved surfaces where improvised explosive devices (IEDs) pose the largest threat (e.g., the exemplary suspension facilitates avoiding enemy engagement, etc.).

Each suspension element of the fully-active and multiple degree of freedom suspension system includes a multiple degree of freedom linkage configured to facilitate increased wheel travel, independent corner control, and active force control, among other features. According to an exemplary embodiment, the suspension elements increase vehicle mobility by substantially increasing suspension travel. The suspension having increased suspension travel facilitates faster vehicle speed on terrain, extreme obstacle negotiation, and increased access to various terrain. By way of example, vehicles equipped with the suspension system of the present disclosure may traverse terrain that is inaccessible by vehicles with other, traditional suspension systems (e.g., trenches, gaps, rubble piles, and rocks may become more negotiable, etc.). By way of another example, vehicles having the exemplary suspension system may have increased vertical suspension travel, be faster, and highly mobile (e.g., over rough terrain, etc.). Greater vertical wheel travel may also improve a vehicle's ride quality compared to traditional vehicles with less vertical wheel travel.

According to an exemplary embodiment, the suspension system includes or is coupled to a control system and an energy storage system. The control system is configured to input energy into the suspension system according to a suspension control strategy and regenerate energy to the vehicle energy storage system during controlled damping. The suspension system may thereby provide both superior ride and improved efficiency compared to passive or semi-active suspension systems. In some embodiments, the suspension system is configured to rapidly lower the vehicle's profile (e.g., center of gravity, signature, etc.) when the vehicle is being engaged by or encountering a threat (e.g., an enemy, a rocket propelled grenade (RPG), a low hanging obstacle, etc.). By reducing its signature (e.g., ducking, squatting, kneeling, etc.), the vehicle may have an increased likelihood of avoiding the threat.

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According to the exemplary embodiment shown in FIGS. 1-3, a suspension system (e.g. a fully-active and multiple degree of freedom suspension system, etc.) includes a plurality of suspension elements, shown as electromechanical suspension elements 100, coupled to a vehicle, shown as vehicle 10. According to an exemplary embodiment, the vehicle 10 is a military ground vehicle. In other embodiments, the vehicle 10 is an off-road vehicle such as an utility task vehicle, a recreational off highway vehicle, an all-terrain vehicle, a sport utility vehicle, and/or still another vehicle. In yet other embodiments, the vehicle 10 is another type of off-road vehicle such as mining, construction, and/or farming equipment. In still other embodiments, the vehicle 10 is an aerial truck, a rescue truck, an aircraft rescue and firefighting (ARFF) truck, a concrete mixer truck, a refuse truck, a commercial truck, a tanker, an ambulance, and/or still another vehicle. The suspension system of the vehicle 10 may be configured for operation on both paved and rough, off-road terrain. As such, the suspension system may be correspondingly configured to support the weight of the vehicle 10 while providing comfortable ride quality on both paved and rough, off-road terrain.

As shown in FIGS. 1-2, the vehicle 10 includes a hull and frame assembly 20 (e.g., a monocoque, etc.). In other embodiments, the vehicle 10 includes a chassis or frame (e.g., frame rails, etc.) that supports a body assembly. As shown in FIGS. 1-2, the hull and frame assembly 20 defines a first longitudinal side, shown as front 22, a second longitudinal side, shown as rear 24, a first lateral side, shown as left side 26, and a second lateral side, shown as right side 28. The vehicle 10 includes a plurality of tractive element assemblies, shown as front tractive element assemblies 40 and rear tractive element assemblies 42. In some embodiments, the vehicle 10 includes a plurality of front tractive element assemblies 40 and/or a plurality of rear tractive element assemblies 42 positioned on each of the left side 26 and right side 28 of the vehicle 10 (e.g., two, three, four, etc.). The front tractive element assemblies 40 and/or the rear tractive element assemblies 42 may include brakes (e.g., disc brakes, drum brakes, air brakes, etc.), gear reductions, steering components, wheel hubs, wheels, and/or other features. As shown in FIGS. 1-2, the front tractive element assemblies 40 and the rear tractive element assemblies 42 include a tractive element, shown as wheel and tire assembly 44. In other embodiments, at least one of the front tractive element assemblies 40 and rear tractive element assemblies 42 include a different type of tractive element (e.g., a track, etc.).

According to an exemplary embodiment, electromechanical suspension element 100 of the suspension system is configured to couple one of the front tractive element assemblies 40 and/or one of the rear tractive element assemblies 42 to the hull and frame assembly 20 (or frame, chassis, body, etc.) of the vehicle 10 or another portion thereof. In one embodiment, the electromechanical suspension elements 100 are configured as leading-arm suspension elements (e.g., where the tractive elements are positioned in front of pivot axes for the respective electromechanical suspension elements 100, etc.). In other embodiments, the electromechanical suspension elements 100 are configured as trailing-arm suspension elements (e.g., where the tractive elements are positioned behind pivot axes for the respective electromechanical suspension elements 100, etc.). In still other embodiments, the electromechanical suspension elements 100 that couple the front tractive element assemblies 40 to the hull and frame assembly 20 are configured as leading-arm suspension elements and the electromechanical

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suspension elements 100 that couple the rear tractive element assemblies 42 to the hull and frame assembly 20 are configured as trailing-arm suspension elements. In an alternative embodiment, only the front tractive element assemblies 40 are coupled to the hull and frame assembly 20 with the electromechanical suspension elements 100 and the rear tractive element assemblies 42 are coupled to the hull and frame assembly 20 with another suspension system (e.g., traditional spring and damper suspension elements, etc.), or vice versa. In another alternative embodiment, the front tractive element assemblies 40 are coupled to the hull and frame assembly 20 with two-degree of freedom electromechanical suspension elements 100 (e.g., facilitate both rotation and extension and retraction, etc.) and the rear tractive element assemblies 42 are coupled to the hull and frame assembly 20 with one-degree of freedom electromechanical suspension elements 100 (e.g., facilitate rotation or extension and retraction, etc.), or vice versa.

As shown in FIG. 2, the vehicle 10 includes a powertrain system, shown as powertrain 50. The powertrain 50 includes a primary driver, shown as engine 52, an energy generation device, shown as generator 54, and an energy storage device (e.g., battery, capacitors, ultra-capacitors, etc.), shown as energy storage device 58, electrically coupled to the generator 54 by a connection (e.g., wire, lead, etc.), shown as electrical connection 56. The engine 52 may receive fuel (e.g., gasoline, diesel, etc.) from a fuel tank and combust the fuel to generate mechanical energy. A transmission may receive the mechanical energy and provides an output to the generator 54. In other embodiments, the engine 52 drives the generator 54 directly. The generator 54 is configured to convert the mechanical energy into electrical energy that may be stored by the energy storage device 58. The energy storage device 58 may provide electrical energy to a motive driver to drive at least one of the front tractive element assemblies 40 and the rear tractive element assemblies 42. In some embodiments, each of the front tractive element assemblies 40 and/or the rear tractive element assemblies 42 include an individual motive driver (e.g., a motor that is electrically coupled to the energy storage device 58, etc.) configured to facilitate independently driving each of the wheel and tire assemblies 44. In an alternative embodiment, a transmission of the vehicle is rotationally coupled to the engine 52 and a drive shaft. The drive shaft may be received by a differential configured to convey the rotational energy of the drive shaft to a final drive (e.g., half-shafts coupled to the wheel and tire assemblies 44, etc.). The final drive then propels or moves the vehicle 10. In such embodiments, the vehicle 10 may not include the generator 54 and/or the energy storage device 58. According to an exemplary embodiment, the engine 52 is a compression-ignition internal combustion engine that utilizes diesel fuel. In alternative embodiments, the engine 52 is another type of device (e.g., spark-ignition engine, fuel cell, electric motor, etc.) that is otherwise powered (e.g., with gasoline, compressed natural gas, hydrogen, electricity, etc.).

As shown in FIG. 2, the vehicle 10 includes one or more steering assemblies, shown as steering systems 60, configured to steer the vehicle 10. The steering systems 60 may include one or more braking systems (e.g., disc brakes, air brakes, drum brakes, etc.), shown as brakes 62, configured to brake the vehicle 10. In other embodiments, the vehicle 10 includes both steering systems 60 and brakes 62. In some embodiments, the vehicle 10 includes one or more front steering systems 60 (e.g., to facilitate steering the front tractive element assemblies 40 based on a steering wheel position, etc.) and/or one or more rear steering systems 60

(e.g., to facilitate steering the rear tractive element assemblies **42** based on a steering wheel position, etc.). In one embodiment, each of the front tractive element assemblies **40** and/or the rear tractive element assemblies **42** include an actuator (e.g., linear actuator, rotational actuator, etc.) electrically coupled to the energy storage device **58** and configured to facilitate independently steering each of the wheel and tire assemblies **44** (e.g., actuating a king pin steering mechanism with the actuator, etc.). In another embodiment, the vehicle **10** is steered using skid steering (e.g., the brakes **62** and/or the motive drivers are used to steer the vehicle **10**, etc.). By way of example, the individual motive drivers and/or the brakes **62** may be used to operate the wheel and tire assemblies **44** at different speeds to steer the vehicle **10**. For example, a front tractive element assembly **40** on the left side **26** of the vehicle **10** may be driven slower (e.g., by changing the speed of individual motive drivers, by engaging a brake **62**, etc.) than a front tractive element assembly **40** on the right side **28** of the vehicle **10** such that the vehicle **10** turns left. In some embodiments, the vehicle **10** is steered cooperatively using skid steering and king pin steering with actuators. In an alternative embodiment, the vehicle **10** includes a traditional steering mechanism (e.g., a rack and pinion steering mechanism with king pin steering, etc.).

As shown in FIGS. 1-2, the vehicle **10** includes a communication system, shown as communication system **70**, first sensors, shown as terrain sensors **80**, and a threat avoidance system, shown as threat avoidance system **90**. The threat avoidance system **90** includes second sensors, shown as threat detection sensors **92**, according to the exemplary embodiment shown in FIGS. 1-2. According to an exemplary embodiment, the communication system **70** is configured to communicate with an external system such as a global positioning system (GPS) and/or a topography system. Communication with a GPS and/or a topography system may facilitate receiving topography data indicative of the topography of a surrounding environment or landscape around and/or ahead of the vehicle **10** (e.g., based on the current position of the vehicle **10**, based on the direction of travel of the vehicle **10**, etc.). According to an exemplary embodiment, the terrain sensors **80** are configured to acquire topography data indicative of the topography of a surrounding environment (e.g., terrain, landscape, trenches, gaps, rubble piles, rocks, etc.) proximate the vehicle **10** (e.g., forward, to the side, to the rear, within the range of the terrain sensors **80**, to facilitate terrain mapping, etc.). The terrain sensors **80** may be variously positioned about the vehicle **10** (e.g., disposed along the front **22**, the rear **24**, the left side **26**, the right side **28**, etc.). According to an exemplary embodiment, the threat detection sensors **92** of the threat avoidance system **90** are configured to acquire data indicative of a potential threat and/or an actual threat near the vehicle **10** (e.g., an incoming RPG, a low hanging obstacle, etc.). The threat detection sensors **92** may be variously positioned about the vehicle **10** (e.g., disposed along the front **22**, the rear **24**, the left side **26**, the right side **28**, etc.). The control system may use the topography data and/or the threat data to control the electromechanical suspension elements **100** of the suspension system.

As shown in FIG. 3, the electromechanical suspension element **100** includes a main body or linkage, shown as housing **110**. The housing **110** defines a longitudinal axis, shown as longitudinal axis **102**, that extends along the length of the housing **110**. The housing **110** has a first end, shown as proximal end **112**, and an opposing second end, shown as distal end **114**. As shown in FIG. 3, the proximal end **112** includes a first coupler, shown as proximal eyelet **116**, and

the distal end **114** includes a second coupler, shown as distal eyelet **118**. According to the exemplary embodiment shown in FIG. 3, the proximal eyelet **116** is configured to facilitate coupling the proximal end **112** of the housing **110** to a portion of the hull and frame assembly **20** (e.g., chassis, side plate, hull, body, bracket, etc.) with a mount, shown as suspension mount **30**, and the distal eyelet **118** is configured to facilitate coupling the distal end **114** of the housing **110** to one of the wheel and tire assemblies **44** (e.g., thereby coupling the wheel and tire assemblies **44** to the hull and frame assembly **20**, etc.). According to an exemplary embodiment, the proximal eyelet **116** and the distal eyelet **118** are integrally formed with the housing **110**.

As shown in FIG. 3, the electromechanical suspension element **100** includes a first joint, shown as linear joint **120**, that provides a first degree of freedom (e.g., linear extension and retraction of the electromechanical suspension element **100**, etc.) and a second joint, shown as rotational joint **130**, that provides a second degree of freedom (e.g., rotational movement of the electromechanical suspension element **100** about a pivot axis extending through the proximal eyelet **116**, etc.). According to an exemplary embodiment, the linear joint **120** and the rotational joint **130** are configured to provide fully-active actuation (e.g., capable of adding energy with a controlled motoring force and removing energy with a controlled damping force, etc.). In other embodiments, at least one of the linear joint **120** and the rotational joint **130** are configured to provide another actuation profile (e.g., semi-active actuation, passive actuation, etc.).

As shown in FIG. 3, the linear joint **120** includes a first actuator housing, shown as linear actuator housing **122**, a first tube, shown as outer tube **124**, and a second tube, shown as inner tube **126**. The inner tube **126** is configured to translate along the longitudinal axis **102** (e.g., into and out of the outer tube **124**, etc.). In one embodiment, the outer tube **124** and the inner tube **126** have circular cross-sections. According to an exemplary embodiment, the outer tube **124** has an inner diameter that is approximately equal to the outer diameter of the inner tube **126** such that the inner tube **126** is received in the outer tube **124**. In other embodiments, the outer tube **124** and the inner tube **126** have another cross-sectional shape (e.g., square, hexagonal, rectangular, oval, etc.). In still other embodiments, the electromechanical suspension element **100** includes another structure configured to facilitate extension and retraction thereof (e.g., a bellows, etc.). As shown in FIG. 3, the outer tube **124** and the inner tube **126** cooperatively define a cavity, shown as internal volume **128**. Movement of the inner tube **126** relative to the outer tube **124** causes the internal volume **128** to increase or decrease in size. By way of example, the internal volume **128** decreases as the inner tube **126** retracts along the longitudinal axis **102** into the outer tube **124** and increases as the inner tube **126** extends along the longitudinal axis **102** out of the outer tube **124**.

As shown in FIG. 3, the linear joint **120** includes a first actuation system, shown as linear actuation system **150**, having a first actuator, shown as linear actuator **152**, and a first resilient member, shown as linear resilient member **154**. The linear actuator housing **122** is configured to receive and store the linear actuator **152**, according to an exemplary embodiment. The outer tube **124** and the inner tube **126** are configured to cooperatively receive and store the linear resilient member **154** within the internal volume **128**.

According to an exemplary embodiment, the linear resilient member **154** is configured to support at least a portion of the static load (e.g., weight, sprung mass, etc.) of the

vehicle 10. In one embodiment, the linear resilient member 154 is or includes a coil spring. In another embodiment, the linear resilient member 154 is or includes a hydro-pneumatic spring. In still another embodiment, the linear resilient member 154 is or includes another type of resilient member capable of supporting the static load of the vehicle 10 (e.g., a gas spring, etc.).

According to an exemplary embodiment, the linear actuator 152 is configured to facilitate linear extension and retraction of the inner tube 126 along the longitudinal axis 102 into and out of the outer tube 124, as indicated by directional arrow 104. The linear actuator 152 is electrically coupled to the energy storage device 58 and configured to extend and/or retract the linear joint 120 (e.g., compress and/or expand the linear resilient member 154, etc.), according to an exemplary embodiment. In other embodiments, the linear actuator 152 is otherwise powered (e.g., pneumatically, hydraulically, etc.). The linear joint 120 extension and retraction provides compliance and facilitates absorbing forces incident in the longitudinal direction of the vehicle 10.

As shown in FIG. 3, the rotational joint 130 includes a second actuator housing, shown as rotational actuator housing 132. According to an exemplary embodiment, the rotational actuator housing 132 is coupled to (e.g., attached to, fixed to, integrally formed with, etc.) the linear actuator housing 122 by an intermediate structural element, shown as intermediate member 140. As shown in FIG. 3, the rotational joint 130 is configured to receive at least one component of a second actuation system, shown as rotational actuation system 160, having a second actuator, shown as rotational actuator 162, and a second resilient member, shown as rotational resilient member 164. The rotational actuator housing 132 is configured to receive and store the rotational actuator 162, according to an exemplary embodiment. In an alternative embodiment, the rotational actuator 162 is coupled to the housing 110 (e.g., to the rotational actuator housing 132 at the proximal eyelet 116, coaxial with a pivot axis through the proximal eyelet 116, coupled along the length of the housing 110, etc.) and positioned externally relative to the rotational actuator housing 132. In one embodiment, the rotational resilient member 164 is rotationally coupled to the housing 110 (e.g., to the rotational actuator housing 132 at the proximal eyelet 116, coaxial with a pivot axis through the proximal eyelet 116, to the proximal end 112 of the housing 110, etc.). In another embodiment, the rotational resilient member 164 is received by and disposed within the rotational actuator housing 132.

According to an exemplary embodiment, the rotational resilient member 164 is configured to support at least a portion of the static load (e.g., weight, sprung mass, etc.) of the vehicle 10. The linear resilient members 154 and the rotational resilient members 164 of the electromechanical suspension elements 100 are configured to cooperatively support the entire static load of the vehicle 10, according to an exemplary embodiment. In one embodiment, the rotational resilient member 164 is or includes a torsion bar. By way of example, a first end of the torsion bar may be coupled (e.g., fixed, welded, fastened, etc.) to a portion of the hull and frame assembly 20 (e.g., chassis, frame, mounting location, bracket, suspension mount 30, etc.), and a second end of the torsion bar may be coupled to the housing 110 of the electromechanical suspension element 100 (e.g., coaxially coupled along a pivot axis of the proximal eyelet 116, coupled to the rotational actuator housing 132, coupled to the linear actuator housing 122, coupled to the intermediate member 140, etc.). In another embodiment, the rotational

resilient member 164 is or includes a bell crank assembly. The bell crank assembly may include a cam having a lobe and a linear resilient element (e.g., a coil spring, etc.). By way of example, the cam may be coaxially and rotationally coupled along the pivot axis of the housing 110 (e.g., an axis through the proximal eyelet 116, etc.). The lobe may be configured to engage the linear resilient element. The linear resilient element may be positioned to bias the cam and the electromechanical suspension element 100 to a static load position (e.g., providing a rotational biasing force, etc.). In other embodiments, the rotational resilient member 164 is or includes a hydro-pneumatic spring. In still other embodiments, the rotational resilient member 164 is or includes a rotational spring element. In yet other embodiments, the rotational resilient member 164 is or includes another type of resilient member capable of supporting the static load of the vehicle 10.

According to an exemplary embodiment, the rotational actuator 162 is configured to rotate the electromechanical suspension element 100 about the proximal eyelet 116 (i.e., about the pivot axis thereof), as indicated by directional arrow 106. The rotational actuator 162 is electrically coupled to the energy storage device 58 and configured to rotate the electromechanical suspension element 100, according to an exemplary embodiment. In other embodiments, the rotational actuator 162 is otherwise powered (e.g., pneumatically, hydraulically, etc.). The electromechanical suspension element 100 rotation provides compliance and facilitates absorbing forces incident in a vertical direction (e.g., by facilitating a relatively large amount of vertical travel of the front tractive element assemblies 40 and/or the rear tractive element assemblies 42, etc.).

According to an exemplary embodiment, the rotational actuator 162 pivots the electromechanical suspension element 100 through a range of motion that corresponds with a vertical displacement of the wheel and tire assembly 44. In one embodiment, the maximum vertical displacement of the wheel and tire assembly 44 is between three and four feet. In other embodiments, the maximum vertical displacement of the wheel and tire assembly 44 is less than or greater than three to four feet. The suspension system having the large amount of suspension travel facilitates increased vehicle speed on terrain, extreme obstacle negotiation, and increased access to various terrain. By way of example, the electromechanical suspension elements 100 may facilitate traversing terrain that is inaccessible by vehicles with other, traditional suspension systems (e.g., trenches, gaps, rubble piles, and rocks may become more negotiable, etc.). By way of another example, the electromechanical suspension elements 100 may facilitate designing fast, highly mobile, off-road and/or on-road vehicles, while improving ride quality.

In some embodiments, the linear actuation system 150 includes a first sensor, shown as linear sensor 156. According to an exemplary embodiment, the linear sensor 156 is positioned to acquire position data indicative of the extension or retraction of the linear joint 120. By way of example, the linear sensor 156 may be positioned to acquire data relating to the position of the inner tube 126 (e.g., relative to a static position or static ride height, relative to the linear sensor 156, based on an amount of actuation of the linear actuator 152, etc.) and/or the amount of actuation of the linear actuator 152. In some embodiments, the rotational actuation system 160 includes a second sensor, shown as rotational sensor 166. According to an exemplary embodiment, the rotational sensor 166 is positioned to acquire position data indicative of the rotational position of the

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electromechanical suspension element **100** (e.g., relative to a static position, relative to the rotational sensor **166**, etc.).

According to the exemplary embodiment shown in FIG. **4**, a control system **400** for a vehicle (e.g., the vehicle **10**, etc.) includes a controller **410**. In one embodiment, the controller **410** is configured to selectively engage, selectively disengage, control, and/or otherwise communicate with components of the vehicle **10**. As shown in FIG. **4**, the controller **410** is coupled to the steering system **60**, a topography system **430** (e.g., with the communication system **70**, etc.), the terrain sensors **80**, the threat avoidance system **90**, the linear actuator **152**, the linear sensor **156**, the rotational actuator **162**, the rotational sensor **166**, and a user interface **420**. In other embodiments, the controller **410** is coupled to more or fewer components. The controller **410** may be configured to selectively control the position (e.g., rotation, extension, retraction, etc.) of the electromechanical suspension elements **100** to facilitate providing a near-constant force to the vehicle **10**. By way of example, the controller **410** may send and receive signals with the steering system **60**, the topography system **430** with the communication system **70**, the terrain sensors **80**, the threat avoidance system **90**, the linear actuator **152**, the linear sensor **156**, the rotational actuator **162**, the rotational sensor **166**, and the user interface **420**.

The controller **410** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital-signal-processor (DSP), circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. According to the exemplary embodiment shown in FIG. **4**, the controller **410** includes a processing circuit **412** and a memory **414**. The processing circuit **412** may include an ASIC, one or more FPGAs, a DSP, circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. In some embodiments, the processing circuit **412** is configured to execute computer code stored in the memory **414** to facilitate the activities described herein. The memory **414** may be any volatile or non-volatile computer-readable storage medium capable of storing data or computer code relating to the activities described herein. According to an exemplary embodiment, the memory **414** includes computer code modules (e.g., executable code, object code, source code, script code, machine code, etc.) configured for execution by the processing circuit **412**. The memory **414** includes various actuation profiles corresponding to loading conditions experienced by the suspension system of the vehicle **10**, according to an exemplary embodiment. In some embodiments, controller **410** represents a collection of processing devices (e.g., servers, data centers, etc.). In such cases, the processing circuit **412** represents the collective processors of the devices, and the memory **414** represents the collective storage devices of the devices.

In one embodiment, the user interface **420** includes a display and an operator input. The display may be configured to display a graphical user interface, an image, an icon, and/or still other information. In one embodiment, the display includes a graphical user interface configured to provide general information about the vehicle **10** (e.g., vehicle speed, fuel level, warning lights, etc.). The graphical user interface may also be configured to display a current position of the vehicle **10**, one or more orientations of the electromechanical suspension elements **100**, potential

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threats or incoming threats (e.g., a RPG, a low hanging obstacle, etc.), a suggested path or route to follow, and/or still other information relating to the vehicle **10** and/or the electromechanical suspension elements **100**.

The operator input may be used by an operator to provide commands to at least one of the steering system **60**, the communication system **70**, the linear actuator **152**, and the rotational actuator **162**. The operator input may include one or more buttons, knobs, touchscreens, switches, levers, joysticks, pedals, a steering wheel, or handles. The operator input may facilitate manual control of some or all aspects of the operation of the electromechanical suspension elements **100** and the vehicle **10**. It should be understood that any type of display or input controls may be implemented with the systems and methods described herein.

The controller **410** may be configured to control the linear actuator **152** and/or the rotational actuator **162** of each of the electromechanical suspension elements **100** to facilitate providing a near-constant force to the hull and frame assembly **20** of the vehicle **10** (e.g., by adjusting the pitch and roll of the vehicle **10**, by absorbing input forces from the terrain or obstacles, etc.). In some embodiments, the controller **410** is configured to provide a command to the linear actuators **152** and/or the rotational actuators **162** positioned on the left side **26** and/or the right side **28** to adjust the roll of the vehicle **10**. In some embodiments, the controller **410** is configured to provide a command to the linear actuators **152** and/or the rotational actuators **162** positioned at the front **22** and/or the rear **24** to adjust the pitch of the vehicle **10**. In some embodiments, the controller **410** is configured to provide a command to any of the linear actuators **152** and/or the rotational actuators **162** to otherwise adjust the orientation of the vehicle **10**.

In some embodiments, the controller **410** is configured to reactively control the linear actuators **152** and/or the rotational actuators **162** based on information feedback from the linear sensor **156** and/or the rotational sensor **166**, respectively. By way of example, the controller **410** may provide a command to at least one of the linear actuator **152** and the rotational actuator **162** to lift (e.g., retract, rotate upward, etc.) a respective wheel and tire assembly **44** over an obstacle (e.g., rock, bump, etc.) in response to the linear sensor **156** and/or the rotational sensor **166** acquiring data that indicates such an obstacle has been encountered. By way of another example, the controller **410** may provide a command to at least one of the linear actuator **152** and the rotational actuator **162** to push (e.g., extend, rotate downward, etc.) a respective wheel and tire assembly **44** into an obstacle (e.g., pothole, trench, etc.) in response to the linear sensor **156** and/or the rotational sensor **166** acquiring data that indicates such an obstacle has been encountered. The controller **410** is thereby capable of adding energy via a controlled motoring force or removing energy via a controlled damping force. The linear sensor **156** and/or the rotational sensor **166** may be or include at least one of position sensors and load sensors, among other types of sensors, positioned to provide signals relating to obstacles and/or the surrounding terrain.

In some embodiments, the controller **410** is configured to actively control the linear actuators **152** and/or the rotational actuators **162** based on information feedback from the topography system **430** and/or the terrain sensors **80**. By way of example, the controller **410** may receive topography data indicative of the topography of a surrounding environment or landscape around and/or ahead of the vehicle **10** (e.g., based on the current position of the vehicle **10**, based on the direction of travel of the vehicle **10**, etc.). By way of

another example, the terrain sensors **80** may be configured to acquire topography data indicative of the topography of a surrounding environment (e.g., terrain, landscape, trenches, gaps, rubble piles, rocks, etc.) around the vehicle **10** (e.g., within the range of the terrain sensors **80**, etc.). The controller **410** may provide a command to at least one of the linear actuators **152** and the rotational actuators **162** based on the topography data to compensate for obstacles and/or changes in the terrain to facilitate providing a near-constant force to the hull and frame assembly **20** (or chassis, frame, and/or body). For example, the controller **410** may be configured to provide a command to the linear actuators **152** and/or the rotational actuators **162** of one or more of the electromechanical suspension elements **100** to rotate, extend, and/or retract the electromechanical suspension elements **100** in order to level the vehicle **10** in response to the topography data indicating that the vehicle **10** is traveling on an uneven surface. In another example, the controller **410** may be configured to provide a command to the linear actuator **152** and/or the rotational actuator **162** of a respective electromechanical suspension element **100** to rotate, extend, and/or retract the electromechanical suspension element **100** in response to the topography data indicating the wheel and tire assembly **44** coupled to the electromechanical suspension element **100** is encountering or about to encounter an obstacle (e.g., a rock, pothole, trench, gap, rubble pile, etc.).

In some embodiments, the controller **410** is configured to actively control the linear actuators **152** and/or the rotational actuators **162** based on data provided by the threat detection sensors **92** of the threat avoidance system **90**. By way of example, the threat detection sensors **92** may be configured to acquire threat data indicative of a potential threat and/or an actual threat near and/or approaching the vehicle **10**. The controller **410** may thereby provide a command to at least one of the linear actuators **152** and the rotational actuators **162** based on the threat data to reduce the likelihood of encountering the threat. For example, the controller **410** may be configured to provide a command to the linear actuators **152** and/or the rotational actuators **162** of one or more of the electromechanical suspension elements **100** to rotate, extend, and/or retract the electromechanical suspension elements **100** in order to reduce the signature (e.g., center of gravity, etc.) of the vehicle **10** (e.g., by kneeling, squatting, ducking, etc.) to avoid a low hanging obstacle (e.g., a tree branch, a low height clearance, etc.) and/or an incoming obstacle (e.g., enemy fire, an RPG, etc.). In another example, the controller **410** is configured to provide a command to the linear actuators **152** and/or the rotational actuators **162** of one or more of the electromechanical suspension elements **100** to rotate, extend, and/or retract the electromechanical suspension elements **100** in order to increase the height of the vehicle **10** to avoid an obstacle on the ground (e.g., a tree stump, a boulder, an IED, etc.) and/or a low incoming obstacle (e.g., enemy fire, an RPG, a grenade, an animal, etc.).

In some embodiments, the controller **410** is configured to accommodate different passive spring rates (e.g., of the linear resilient member **154**, the rotational resilient member **164**, etc.), load fluctuations (e.g., from additional passengers or cargo, encountering obstacles, etc.), center of gravity shifts of the vehicle **10** (e.g., from additional passengers, cargo, etc.), etc. In some embodiments, the controller **410** is configured to control the linear actuators **152** and/or the rotational actuators **162** to isolate the hull and frame assembly **20** from vibrations (e.g., onboard the vehicle **10**, incident upon the vehicle **10**, etc.). In some embodiments, the con-

troller **410** is configured to control the linear actuators **152** and/or the rotational actuators **162** to maintain the vehicle **10** at a target height (e.g., a static loading height, etc.) in response to different load conditions (e.g., additional weight, etc.).

Orbital Wheel Drive

Overview

Referring particularly to FIGS. **5** and **6**, a suspension system **500** includes an attachment member, a control arm, a frame member, etc., shown as control arm **502**. The suspension system **500** also includes a hub assembly, a wheel attachment member, etc., shown as wheel hub **580** and a power transmission system **570** configured to transfer power from the control arm **502** to the wheel hub **580**. The power transmission system **570** includes multiple gear systems, gear trains, etc. Specifically, the power transmission system **570** includes a first or drive gear train **572**, a second or steering gear train **574**, and a third or steering knuckle gear train **576**.

Drive Gear Train

The drive gear train **572** is configured to transfer power to a wheel, a tractive member, a tractive element, a rotational member, a driven member, etc., shown as tractive element **504**. The drive gear train **572** is configured to receive input power (e.g., rotational kinetic energy, torque, rotation, etc.) through an input shaft **510**. The input shaft **510** can be driven to rotate about an axis **512** that extends centrally through the input shaft **510**. The drive gear train **572** outputs torque or power (e.g., rotational kinetic energy) to the tractive element **504** through an output shaft **532**. An axis **528** extends centrally and longitudinally through the output shaft **532** and is laterally offset from the axis **512**. In this way, the drive gear train **572** transfers rotational kinetic energy from the input shaft **510** to the output shaft **532** that are laterally offset from each other.

The input shaft **510** includes a spline, a geared end, etc., shown as teeth **514**. The input shaft **510** is configured to receive rotational kinetic energy through the teeth **514**. For example, the input shaft **510** can be driven to rotate through the teeth **514** and a member (e.g., a female member) that is configured to engage the input shaft **510** through the teeth **514**. The drive gear train **572** includes a first gear **516** that is fixedly and rotatably coupled with the input shaft **510**. The first gear **516** includes teeth **518** and is configured to engage and drive a laterally offset gear, shown as second gear **524**. The second gear **524** is fixedly and rotatably coupled with an intermediate shaft **522** that is laterally offset from the input shaft **510**. An axis **520** extends centrally through the intermediate shaft **522** and is laterally offset from the axis **512**. The intermediate shaft **522** includes the second gear **524** and a third gear **526** positioned or disposed at opposite ends of the intermediate shaft **522**. The second gear **524** and the third gear **526** can be similar to the first gear **516** and can include similar teeth. In some embodiments, the second gear **524** and the third gear **526** have a same number of teeth. In other embodiments, the second gear **524** and the third gear **526** have a different number of teeth (e.g., the second gear **524** has more teeth than the third gear **526** or vice versa). The second gear **524** and the third gear **526** are longitudinally offset at opposite ends of the intermediate shaft **522**.

The third gear **526** is configured to engage a corresponding gear, shown as fourth gear **530**. The fourth gear **530** is fixedly and rotatably coupled with an output shaft **532** such that rotation of the third gear **526** drives the output shaft **532** to also rotate. The output shaft **532** is laterally offset from both the axis **520** that extends through the intermediate shaft **522** and the axis **512** that extends through the input shaft

510. The output shaft **532** includes the fourth gear **530** positioned at a first end, and an engagement portion, teeth, a spline, etc., shown as engagement portion **586**. The engagement portion **586** is configured to engage a corresponding portion of the tractive element **504** to drive the tractive element **504**. In this way, rotational kinetic energy is transferred through the drive gear train **572** from the input shaft **510**, through the first gear **516**, the second gear **524**, the intermediate shaft **522**, the third gear **526**, the fourth gear **530**, and the output shaft **532** to the tractive element **504**. The drive gear train **572** can receive input rotational kinetic energy at a speed ω_1 with a torque T_1 and may output kinetic energy at a speed ω_2 with a torque T_2 . In some embodiments, the output torque T_2 is greater than the input torque T_1 (e.g., $T_2 > T_1$) and the output speed ω_2 is less than the input speed ω_1 (e.g., $\omega_2 < \omega_1$). In other embodiments, the output torque T_2 is less than the input torque T_1 (e.g., $T_2 < T_1$) and the output speed ω_2 is greater than the input speed ω_1 (e.g., $\omega_2 > \omega_1$). In other embodiments, the output torque T_2 is equal to the input torque T_1 (e.g., $T_2 = T_1$) and the output speed ω_2 is equal to the input speed ω_1 (e.g., $\omega_2 = \omega_1$).

The output shaft **532** can be configured to directly engage the tractive element **504** or may be configured to indirectly engage and drive the tractive element **504**. For example, the output shaft **532** can be configured to drive an axle (e.g., a constant velocity joint) disposed between the output shaft **532** and the tractive element **504**. In this way, an orientation of the tractive element **504** (e.g., a steering orientation) may change, while still facilitating the transmission of drive power to the tractive element **504**.

Steering Gear Train

Referring still to FIGS. **5** and **6**, the steering gear train **574** includes a steering gear, shown as first gear **540** and a steering shaft, shown as first shaft **542**. The first gear **540** and the first shaft **542** include an inner volume through which the input shaft **510** extends. The input shaft **510** may slidably couple with or slidably engage an inner surface of the first gear **540** and the input steering shaft **542**. For example, an inner surface of the first gear **540** and the input steering shaft **542** can be configured to slidably engage an outer surface of the input shaft **510**. The first gear **540** and the input steering shaft **542** can be co-axial with the input shaft **510**.

The first gear **540** is configured to be driven by an input gear **538** and an input shaft **536**. The input gear **538** is fixedly and rotatably coupled with the input shaft **536**. The input shaft **536** and the input gear **538** are laterally offset from the input shaft **510** of the drive gear train **572** and the first shaft **542** of the steering gear train **574**. The input gear **538** and the input shaft **536** can receive input rotational kinetic energy and transfer the rotational kinetic energy to the first gear **540** and the first shaft **542**.

The steering gear train **574** includes a first gear **544** that is fixedly and rotatably coupled with the first shaft **542**. The first gear **544** is driven by the first shaft **542** and transfers rotational kinetic energy to a second gear **546** of the steering gear train **574**. The first gear **544** can be slidably coupled with the exterior surface of the input shaft **510** such that the input shaft **510** and the gear **544** can rotate independently of each other. The first gear **544** is configured to drive a second gear **546**. The first gear **544** and the second gear **546** can mesh or engage each other (e.g., through correspondingly configured teeth) to facilitate the transfer of rotational kinetic energy from the first gear **544** to the second gear **546**. The second gear **546** can be slidably coupled with the intermediate shaft **522** such that the second gear **546** can rotate independently of the intermediate shaft **522**. The

second gear **546** of the steering gear train **574** can be co-axial with the intermediate shaft **522** of the drive gear train **572**. The intermediate shaft **522** can extend through the second gear **546** (e.g., through a central aperture of the second gear **546**) such that the intermediate shaft **522** can rotate independently of the second gear **546**. The second gear **546** of the steering gear train **574** can be positioned between the second gear **524** and the third gear **526** of the drive gear train **572**.

The second gear **546** is configured to mesh with or engage a third gear **548** and drive the third gear **548**. The third gear **548** can have correspondingly configured teeth that engage the second gear **546** to facilitate the transfer of rotational kinetic energy from the second gear **546** to the third gear **548**. The third gear **548** may be fixedly coupled or integrally formed with an output shaft **550**. The output shaft **550** can be co-axial with the output shaft **532** of the drive gear train **572**. The output shaft **550** may include an inner volume that is configured to receive the output shaft **532** of the drive gear train **572** therethrough such that the output shaft **550** can rotate relative to the output shaft **532** of the drive gear train **572**.

The output shaft **550** includes an engagement portion, a driving portion, a steering portion, a steering guide, etc., shown as steering member **588**. The steering member **588** can protrude radially outwards from the output shaft **550** and is configured to engage and drive/steer the tractive element **504**. The steering member **588** is configured to engage a steering bevel (e.g., engagement portion **586**) of the wheel hub **580**. In particular, the steering member **588** is configured to drive a first yoke **582** of the wheel hub **580** that the tractive element **504** rotatably couples with. Rotation of the output shaft **550** drives rotation of the first yoke **582** relative to a second yoke **584** of the wheel hub **580** about axis **590**. In this way, the steering gear train **574** can be operated to rotate the tractive element **504** for steering purposes (e.g., to rotate the first yoke **582** relative to the second yoke **584** about the axis **590**). The steering gear train **574** can be driven to rotate the first yoke **582** relative to the second yoke **584** in either direction.

Knuckle Gear Train

Referring still to FIGS. **5-6**, the steering knuckle gear train **576** includes a first gear **554** that is configured to be driven by an input shaft **553**. The input shaft **553** and the first gear **554** are centered about the axis **512**. The input shaft **553** may be co-cylindrical with the first shaft **542** of the steering gear train **574** and the input shaft **510** of the drive gear train **572**. For example, the input shaft **510** of the drive gear train **572** may extend through and be slidably coupled with the first shaft **542** of the steering gear train **574**, with the first shaft **542** of the steering gear train **574** extending through and slidably coupling with the input shaft **553**. The input shaft **553** may fixedly couple and/or rotatably couple with the first gear **554** such that rotational kinetic energy or torque may be input to the first gear **554** through the input shaft **553**.

The first gear **554** can be rotatably and/or slidably coupled with an exterior (e.g., a radially outwards facing surface) of the first shaft **542** such that the first gear **554** can rotate independently of the first shaft **542**. The first gear **554** is configured to engage a second gear **556** of the steering knuckle gear train **576**. The second gear **556** can be rotatably coupled and/or slidably coupled with the intermediate shaft **522** such that the second gear **556** can rotate relative to or independently of the intermediate shaft **522**. For example, the second gear **556** can be rotatably coupled with the intermediate shaft **522** using a bearing, a ball bearing, a roller bearing, a slidable coupling, a low-frictional interface,

etc. The second gear **556** rotates about the axis **520** independently of rotation of the intermediate shaft **522**.

The second gear **556** is configured to drive or engage a third gear **558** which is fixedly coupled with an output shaft **560** of the steering knuckle gear train **576**. The third gear **558** is fixedly coupled or integrally formed with the output shaft **560** such that rotation of the third gear **558** results in rotation of the output shaft **560**. The output shaft **560** and the third gear **558** are rotatably coupled with the output shaft **532** of the drive gear train **572** such that the output shaft **560** and the third gear **558** can rotate independently of the output shaft **532**. The output shaft **550** of the steering gear train **574**, the output shaft **560** of the steering knuckle gear train **576**, and the output shaft **532** of the drive gear train **572** are all co-cylindrical with each other (e.g., centered about the axis **528**).

The output shaft **560** of the steering knuckle gear train **576** is fixedly coupled or integrally formed with the second yoke **584** such that rotation of the output shaft **560** about the axis **528** drives the second yoke **584** to rotate about the axis **528**. In this way, the second yoke **584** can be rotated about the axis **528** by operation (e.g., by driving) the steering knuckle gear train **576**. In some embodiments, rotational kinetic energy can be input into the steering knuckle gear train **576** in either direction to drive the second yoke **584** to rotate about the axis **528** in either direction. For example, the input shaft **553** can be driven to rotate about the axis **512** in either direction, thereby driving the second yoke **584** and the first yoke **582** to rotate about the axis **528** in either direction.

In this way, the steering knuckle gear train **576** can be operated to achieve adjustment or rotation of the wheel hub **580** about the axis **528**, while the steering gear train **574** can be operated to achieve adjustment or rotation of the tractive element **504** about the axis **590**. Advantageously, the power transmission system **570** can be used to adjust an orientation of the tractive element **504** in several directions (e.g., about the axis **590** and the axis **528**) and to transmit power (e.g., rotational kinetic energy) from a primary mover (e.g., an engine, a motor, etc.) of a vehicle to the tractive element **504**.

Referring now to FIGS. 5-8, the suspension system **500** can include three power transmission systems **570** that are all driven by a common input shaft **510**, a common first gear **540**, and a common input shaft **553**. Each of the power transmission systems **570a-c** are configured to transmit power to, steer, and adjust the orientation of the wheel hub **580** for a corresponding wheel hub **580** and a corresponding tractive element **504**. For example, the power transmission system **570a** is configured to provide power to a tractive element **504a** of a wheel hub **580a**. Likewise, the power transmission system **570b** and the power transmission system **570c** are configured to provide rotational power to tractive elements **504b** and **504c** of the wheel hubs **580b** and **580c**, respectively. The power transmission systems **570a-c** can also be operated to rotate the tractive elements **504** about their respective axes **590** as well as about their respective axes **528**. In this way, the tractive elements **504** can each have several degrees of freedom and are able to rotate about the corresponding axis **590** and the axis **528**. In some embodiments, the power transmission systems **570** are configured to rotate in unison about a central axis **512** that extends longitudinally through a center of the suspension system **500**. In some embodiments, the central axis **512** is the axis **512** that extends centrally through the input shaft **510**.

In some embodiments, the control arm **502** includes a first adjustment member **596a** and a second adjustment member **596b**. The first adjustment member **596a** and second adjust-

ment member **596b** can be screws, elongated members, cylindrical members, threaded members, worms, etc. The first adjustment member **596a** and the second adjustment member **596b** are configured to pivotally or rotatably couple with the control arm **502** to facilitate rotation of the power transmission systems **570** and the tractive elements **504** about the axis **512**. In some embodiments, the first adjustment member **596a** and the second adjustment member **596b** are configured to rotate within a corresponding guide structure **594**. For example, the first adjustment member **596a** can be received within and rotatably couple with the guide structure **594a**. The guide structure **594a** can be a hollow cylindrical member that is integrally formed with the control arm **502**. The second adjustment member **596b** is similar to the first adjustment member **596a** and is received within a guide structure **594b** that is positioned on an opposite side of the control arm.

The control arm **502** includes an inner volume **589** and is configured to rotatably couple with a housing member **597** that the power transmission systems **570** extend through. In some embodiments, the housing member **597** includes side walls that are configured to enclose and/or protect the various gears and shafts of the power transmission systems **570**. The housing member **597** includes a male pivotal member **591**. The control arm **502** includes a female pivotal member **598** that is integrally formed with the control arm **502** and is configured to receive the male pivotal member **591** therewithin. The male pivotal member **591** can have a generally cylindrical or rotor shape, and the female pivotal member **598** includes an internal volume that is configured to receive the male pivotal member **591**. For example the male pivotal member **591** can include a radial protrusion that is configured to rotatably or pivotally couple with the female pivotal member **598**. The radial protrusion can include teeth **587** that are configured to engage the threads **595** of the first adjustment member **596a** and the second adjustment member **596b**. The female pivotal member **598** and the male pivotal member **591** can be co-axial with each other and may both be centered about the axis **512**. The female pivotal member **598** and the male pivotal member **591** each include a central opening, a central aperture, an inner volume, etc., through which the first shaft **542** of the steering gear train **574**, the input shaft **553** of the steering knuckle gear train **576**, and the input shaft **510** of the drive gear train **572** extend.

The housing member **597** and thereby the power transmission systems **570** can be driven to rotate about the central axis **512** by the first adjustment member **596a** and the second adjustment member **596b**. For example, the first adjustment member **596a** and the second adjustment member **596b** can be driven to rotate about their respective axes **592** that extend longitudinally through the adjustment members **602**. In some embodiments, the first adjustment member **596a** is driven to rotate about the axis **592a** in a first direction (e.g., a clockwise direction) while the second adjustment member **596b** is driven to rotate about the axis **592b** in a second direction that is opposite the first direction (e.g., a counter-clockwise direction) to drive the male pivotal member **591**, the housing member **597**, and the power transmission systems **570** to rotate about the central axis **512**. The first adjustment member **596a** and the second adjustment member **596b** can be driven to rotate about their respective axes **592** to rotate the power transmission systems **570**, the wheel hubs **580**, and the tractive elements **504** about the central axis **512**.

In this way, the tractive elements **504** can be orbitally rotated. The tractive elements **504** can be rotated in unison

about the central axis **512**. The tractive elements **504** can each receive rotational kinetic energy or mechanical energy to drive the tractive elements **504** through their corresponding power transmission systems **570**. Specifically, each of the tractive elements **504** can be driven to rotate for transportation through the corresponding drive gear trains **572**. Likewise, each of the tractive elements **504** can be driven to rotate for steering through their corresponding steering gear trains **574**. The tractive elements **504** can be rotated about their corresponding axes **528** through their corresponding knuckle gear trains **576**.

Referring particularly to FIG. 9, the suspension system **500** can be configured for two tractive elements **504**. Specifically, the tractive elements **504** can be configured to orbitally rotate in unison about the axis **512**. The tractive elements **504** can be independently or in unison be driven to rotate for transportation through their corresponding drive gear trains **572**, for steering through their corresponding steering gear trains **574**, and can be driven to rotate about their respective axes **512** through their corresponding or respective knuckle gear trains **576**.

Extendable Orbital Suspension System

Referring particularly to FIGS. 10-12, a suspension system **600** for a tractive element **604** includes a power transmission assembly **606**. The power transmission assembly **606** can be configured to provide mechanical energy (e.g., rotational kinetic energy, power, torque, etc.) for transportation. In some embodiments, the power transmission assembly **606** is configured to transfer mechanical energy to the tractive element **604** to rotate the tractive element **604** in direction **650** or to rotate the tractive element **604** in a direction that is opposite the direction **650**.

The tractive element **604** can be the same as or similar to the tractive elements **504**. In some embodiments, the tractive element **604** is a wheel, a tire, a tread, etc., that is configured to frictionally engage a ground surface (e.g., in static friction) and propel a vehicle along the ground surface.

Referring particularly to FIG. 10, the power transmission assembly **606** is configured to rotatably couple with the tractive element **604**. The tractive element **604** may be supported by and able to rotate relative to the power transmission assembly **606**. The power transmission assembly **606** can be configured to drive or transfer mechanical energy to the tractive element **604** to rotate for steering. The power transmission assembly **606** can also be configured to drive or transfer mechanical energy to the tractive element **604** to rotate a wheel hub of the tractive element **604**.

The power transmission assembly **606** can include a plurality of or multiple telescoping cylinders that are configured to slidably translate relative to each other. In some embodiments, the power transmission assembly **606** includes a first outer member, a first telescoping member, a first outer housing, etc., shown as first outer member **608a**. The power transmission assembly **606** can also include a second outer member **608b** or a central outer member **608c**. The power transmission assembly **606** can also include a third outer member, a third telescoping member, a third outer housing member, etc., shown as third outer member **608c**. The first outer member **608a** and the third outer member **608c** can be positioned on either side of the second outer member **608b**. The first outer member **608a**, the second outer member **608b**, and the third outer member **608c** may be hollow members configured to receive and slidably translate relative to corresponding inner members. For example, the first outer member **608a** and the third outer member **608c** can be configured to receive a corresponding first inner member **610a** and a third inner member **610c**, respectively

such that the inner members **610** can translate (e.g., slidably translate) relative to the corresponding outer members **608** (e.g., translate within).

Likewise, the second outer member **608b** is configured to slidably engage, slidably couple with, translatably couples with, etc., a corresponding second or central inner member **610b**. In some embodiments, the first outer member **608a**, the second outer member **608b**, and the third outer member **608c** are integrally formed with each other. In some embodiments, the first inner member **610a**, the second inner member **610b**, and the third inner member **610c** are integrally formed with each other. In some embodiments, the second outer member **608b** and the second inner member **610b** define a central telescoping member, a central telescoping shaft, a central telescoping housing member, etc. The central telescoping member can be supported on either side by the outer members **608a** and **608c** and the corresponding inner members **610a** and **610c**.

In some embodiments, the outer members **608a** and **608c** and their corresponding inner members **610a** and **610c** are suspension components (e.g., damper piston **609a** and damper piston **609c**, respectively). For example, the outer member **608a** and the corresponding inner member **610c** can be a damper or a piston (e.g., damper pistons **609**) that is configured to absorb shocks, impacts, sudden translation of the tractive element **604**, etc., that is transferred to the power transmission assembly **606** through the tractive element **604**. In some embodiments, the dampers or pistons are positioned on either side of the central telescoping member as shown in FIGS. 10-12.

Referring still to FIG. 10, the outer members **608** and their corresponding inner members **610** each define a corresponding longitudinal axis **612**. Specifically, the first outer member **608a** and the first inner member **610a** may define a longitudinal axis **612a**, the second or central outer member **608b** and the corresponding second inner member **610b** may define a longitudinal axis **612b**, and the third outer member **608c** and the corresponding inner member **610c** may define a longitudinal axis **612c**. In some embodiments, the longitudinal axes **612a**-**612c** are parallel with each other.

Referring still to FIG. 10, the power transmission assembly **606** includes a drive train **672** that is configured to receive rotational kinetic energy or mechanical energy and transfer the mechanical energy to the tractive element **604** for transportation purposes (e.g., to rotate the tractive element **604** to transport the vehicle). The drive train **672** includes a central shaft, an input shaft, an elongated member, etc., shown as drive shaft **614**. The drive shaft **614** extends through the second or central outer member **608b** and the second or central inner member **610b**. The drive shaft **614** is configured to receive rotational kinetic energy or mechanical energy and transfer the mechanical energy to the tractive element **604** for transportation. The drive shaft **614** can be a telescoping member, including an outer member **616** and an inner member **618**. The outer member **616** is configured to rotatably couple with the inner member **618** such that the outer member **616** and the inner member **618** rotate in unison. In some embodiments, the outer member **616** includes an inner volume and a spline or interlocking portions that are configured to engage corresponding spline or interlocking portions of the inner member **618**. In some embodiments, the outer member **616** is configured to translate relative to the inner member **618** or vice versa, such that an overall longitudinal length of the outer member **616** and the inner member **618** may increase or decrease. Likewise, the outer member **616** is configured to rotatably couple with

the inner member **618** such that the outer member **616** can drive the inner member **618** to rotate.

The drive shaft **614** includes or is fixedly coupled with an input gear, a first bevel gear, etc., shown as input gear **626**. In some embodiments, the input gear **626** is fixedly coupled with the outer member **616** such that the input gear **626** can be driven to rotate the outer member **616** and the inner member **618**. The input gear **626** can be positioned at a first or proximate end of the outer member **616** that is opposite a second or distal end of the outer member **616** that receives the inner member **618** therewithin.

Referring still to FIG. **10**, the drive shaft **614** includes an output gear, a second bevel gear, etc., shown as output gear **620**. The output gear **620** can be fixedly coupled with the inner member **618**. In some embodiments, the output gear **620** is fixedly and rotatably coupled with the inner member **618** such that rotation of the inner member **618** drives the output gear **620** to rotate. For example, the output gear **620** may be press fit, keyed, fastened, etc., or otherwise fixedly coupled with the inner member **618**. The input gear **626** can be similarly fixedly coupled with the outer member **616**. In some embodiments, the output gear **620** and the input gear **626** are fixedly coupled with the drive shaft **614** at opposite ends of the drive shaft **614**. For example, the output gear **620** can be fixedly coupled with the inner member **618** at an end of the inner member **618** that is opposite the end of the inner member **618** that extends into and slidably couples with the outer member **616**.

The drive train **672** includes a driven gear, a third bevel gear, etc., shown as driven gear **622**. The driven gear **622** is configured to engage, mesh with, etc., the output gear **620** such that the mechanical energy of the output gear **620** is transferred to the driven gear **622** for transportation. The driven gear **622** can be mounted on, fixedly coupled with, press fit onto, keyed with, etc., a corresponding shaft. The shaft may fixedly and rotatably couple with an axle that drives the tractive element **604**. In some embodiments, the output gear **620** and the driven gear **622** are bevel gears that transfer the mechanical energy from a first axis (i.e., the longitudinal axis **612b** that extends through the drive shaft **614**) to a second axis that extends through the driven gear **622**, shown as axis **686**. In some embodiments, the axis **686** is perpendicular with the longitudinal axis **612b**. The driven gear **622** can be rotatably fixedly coupled with an axle **624** that transfers the torque or the rotational kinetic energy to the tractive element **604**.

Referring still to FIG. **10**, the power transmission assembly **606** includes a steering train **674** that is configured to rotate the tractive element **604** for steering. The steering train **674** includes a steering input shaft **630** that is configured to rotate about the longitudinal axis **612b**. The steering input shaft **630** can be co-cylindrical with the drive shaft **614**, or may be co-axial with the longitudinal axis **612b**. The steering input shaft **630** may be hollow and can include an inner volume through which the drive shaft **614** extends. In some embodiments, an exterior surface (e.g., a radially outwards facing surface) of the drive shaft **614** is slidably coupled with an interior surface (e.g., a radially inwards facing surface) of the steering input shaft **630**. The steering input shaft **630** and the drive shaft **614** can be rotatably coupled with each other such that the steering input shaft **630** and the drive shaft **614** can rotate relative to each other and/or independently of each other.

The steering input shaft **630** can be a telescoping member including an inner member **634** and an outer member **632**. The inner member **634** can be a hollow cylindrical and/or elongated member that is configured to be received within a

corresponding inner volume or hollow portion of the outer member **632**. The inner member **634** and the outer member **632** can both be hollow members such that the drive shaft **614** may extend therethrough. A radially outwards facing surface of the inner member **634** can slidably couple with a radially inwards facing surface of the outer member **632**. The inner member **634** and the outer member **632** can be fixedly rotatably coupled (e.g., with a spline or an interlocking fit) with each other such that rotation of the inner member **634** drives rotation of the outer member **632** and vice versa. While the inner member **634** and the outer member **632** are fixedly rotatably coupled, the inner member **634** and the outer member **632** may still be translationally slidably coupled such that the inner member **634** can extend or retract relative to the outer member **632** (or vice versa) thereby increasing or decreasing an overall longitudinal length of the steering input shaft **630**.

Referring still to FIG. **10**, the steering input shaft **630** includes an input gear, an input steering gear, a bevel gear, etc., shown as input steering gear **628**, according to an exemplary embodiment. The input steering gear **628** can be rotatably fixedly coupled with the outer member **632**. Specifically, the input steering gear **628** can be press fit, slip fit, keyed, etc., onto the outer member **632**. In other embodiments, the input steering gear **628** is integrally formed with the outer member **632**. The input steering gear **628** can be longitudinally positioned at an input end of the steering input shaft **630**. For example, the input steering gear **628** can be longitudinally positioned proximate the input gear **626**.

The steering input shaft **630** also includes an output gear, an output steering gear, another bevel gear, etc., shown as output steering gear **636**. The output steering gear **636** can be positioned longitudinally at an opposite end of the steering input shaft **630** as compared to the input steering gear **628**. The input steering gear **628** can be positioned at a first longitudinal end of the steering input shaft **630** (e.g., at the input end) while the output steering gear **636** can be positioned at an opposite end of the steering input shaft **630** (e.g., at an output end, proximate the tractive element **604**).

The output steering gear **636** can be fixedly rotatably coupled with the inner member **634** of the steering input shaft **630**. For example, the output steering gear **636** can be integrally formed, keyed, press fit, etc., onto the inner member **634** of the steering input shaft **630**.

The output steering gear **636** may be a bevel gear that is configured to engage a corresponding steering bevel gear **638**. The steering bevel gear **638** can be configured to drive a steering member that engages a steering bevel. For example, the output steering gear **636** can transfer rotational kinetic energy or torque from the steering input shaft **630** to steer the tractive element **604**. The output steering gear **636** can be configured to rotate to drive an inner yoke **682** to rotate about axis **688** relative to an outer yoke **684**. The inner yoke **682** and the outer yoke **684** may define a hub assembly **690**. The hub assembly **690** can be the same as or similar to the wheel hub **580**. In this way, the outer yoke **684** can be the same as or similar to second yoke **584**, and the inner yoke **682** can be the same as or similar to the first yoke **582**. The hub assembly **690** can include similar configurations to the wheel hub **580** for steering, with the input torque for steering received at the hub assembly **690** from the steering input shaft **630** and the steering bevel gear **638**. In this way, the steering train **674** can be operated independently of the drive train **672** to steer the tractive element **604** (e.g., to rotate the tractive element **604** about the axis **688** in either direction).

Referring still to FIG. 10, the power transmission assembly 606 includes a knuckle drive train 676. The knuckle drive train 676 can be configured to drive the outer yoke 684 to rotate about the axis 686. The knuckle drive train 676 includes a knuckle input shaft 642 that extends longitudinally along the longitudinal axis 612b. The knuckle input shaft 642 can be co-cylindrical with the steering input shaft 630 and the drive shaft 614. The knuckle input shaft 642 can also be a hollow shaft and may include an inner volume through which the steering input shaft 630 and the drive shaft 614 extend. In some embodiments, the knuckle input shaft 642 is a radially outermost shaft of the knuckle input shaft 642, the steering input shaft 630, and the drive shaft 614. For example, the drive shaft 614 may be a central shaft that extends through the steering input shaft 630 (e.g., through an inner cavity or hollow cylindrical portion of the steering input shaft 630), and the steering input shaft 630 may extend through an inner volume, an inner cavity, a hollow portion, etc., of the knuckle input shaft 642. The knuckle input shaft 642 can be rotatably slidably coupled with the steering input shaft 630 such that the knuckle input shaft 642 can rotate about the longitudinal axis 612b independently of the steering input shaft 630. In this way, the knuckle input shaft 642, the steering input shaft 630, and the drive shaft 614 can all be co-cylindrical with each other and rotatably free relative to each other, such that the knuckle input shaft 642, the steering input shaft 630, and the drive shaft 614 can each be driven independently.

The knuckle input shaft 642 can also be a telescoping shaft or an extendable/retractable shaft. The knuckle input shaft 642 includes an outer shaft member 644 and an inner shaft member 646. The inner shaft member 646 can be rotatably fixedly coupled with the outer shaft member 644 and may be received within the outer shaft member 644. For example, the inner shaft member 646 and the outer shaft member 644 can be rotatably fixedly coupled with each other with a spline fit. The inner shaft member 646 and the outer shaft member 644 can be translatably or slidably coupled with each other such that the inner shaft member 646 can translate relative to the outer shaft member 644 in the longitudinal direction (e.g., such that an overall longitudinal length of the knuckle input shaft 642 can increase or decrease). In this way, the inner shaft member 646 and the outer shaft member 644 can be telescoping tubular or shaft member that are rotatably fixedly coupled with each other.

The knuckle input shaft 642 include an input bevel gear, a knuckle input gear, an input gear, etc., shown as knuckle input gear 640. The knuckle input gear 640 can be a bevel gear and may be fixedly coupled with the knuckle input shaft 642 at a first end, an input end, etc., of the knuckle input shaft 642. Specifically, the knuckle input gear 640 can be positioned at an end of the knuckle input shaft 642 that is proximate the input gear 626. The knuckle input gear 640 can be fixedly rotatably coupled with the outer shaft member 644 of the knuckle input shaft 642. The knuckle input gear 640 can be integrally formed, keyed, press fit, etc., or otherwise fixedly coupled with the outer shaft member 644 of the knuckle input shaft 642. The knuckle input gear 640 can be directly adjacent the input steering gear 628 and may be slidably coupled with the input steering gear 628 such that the knuckle input gear 640 can rotate relative to the input steering gear 628.

The knuckle input shaft 642 includes an output gear, a bevel gear, a knuckle output gear, etc., shown as knuckle output gear 648. The knuckle output gear 648 can be longitudinally positioned at an end of the knuckle input shaft 642 that is opposite the knuckle input gear 640. For example,

the knuckle output gear 648 can be longitudinally positioned at an end of the knuckle input shaft 642 that is proximate the tractive element 604. The knuckle output gear 648 can be fixedly coupled (e.g., keyed, press fit, integrally formed, etc.) with the inner shaft member 646. In some embodiments, the knuckle output gear 648 is directly adjacent or engages the output steering gear 636. The knuckle output gear 648 can be slidably and/or rotatably coupled with the output steering gear 636 such that the knuckle output gear 648 may rotate freely or independently relative to the output steering gear 636.

The knuckle output gear 648 is configured to mesh, engage, etc., a corresponding portion of the outer yoke 684 such that rotation of the knuckle output gear 648 drives rotation of the outer yoke 684 about the axis 686. The outer yoke 684 can include teeth that are configured to mesh with or engage teeth of the knuckle output gear 648.

Referring particularly to FIGS. 11-12, the suspension system 600 can include a first power transmitting shaft 702 that is configured to provide the drive shaft 614 with rotational kinetic energy or torque. A lateral axis 712 extends longitudinally through the first power transmitting shaft 702. The lateral axis 712 can extend radially through the longitudinal axis 612b and may be substantially perpendicular with the longitudinal axis 612b. The lateral axis 712 can be a longitudinal axis of the first power transmitting shaft 702.

The first power transmitting shaft 702 can be fixedly coupled with a first bevel gear 704 that is configured to mesh with the input gear 626. The first bevel gear 704 and the input gear 626 may both be bevel gears (e.g., bevel spur gears, bevel helical gears, etc.) such that rotation of the first power transmitting shaft 702 about the lateral axis 712 drives the drive shaft 614 to rotate about the longitudinal axis 612b. The first power transmitting shaft 702 can be driven by an engine, an electric motor, etc. In this way, the first power transmitting shaft 702 can be driven to drive the tractive element 604 for transportation.

The suspension system 600 can include a second power transmitting shaft 706 that is co-cylindrical with the first power transmitting shaft 702 about the lateral axis 712. The second power transmitting shaft 706 can be slidably coupled with the first power transmitting shaft 702 and may include an inner volume, an inner cavity, etc., through which the first power transmitting shaft 702 extends. The second power transmitting shaft 706 can be driven independently of the first power transmitting shaft 702. The second power transmitting shaft 706 is fixedly coupled or integrally formed with a second bevel gear 708. The second bevel gear 708 is configured to engage or mesh with the input steering gear 628 to transfer rotational kinetic energy, mechanical energy, torque, etc., from the second power transmitting shaft 706 to the input steering gear 628 (e.g., for steering the tractive element 604). In this way, the tractive element 604 can be steered by driving the second power transmitting shaft 706 independently of the first power transmitting shaft 702.

Referring still to FIGS. 11-12, the suspension system 600 includes a third bevel gear 710 that is configured to engage the knuckle input gear 640. The third bevel gear 710 can be rotatably coupled with the second power transmitting shaft 706 such that the third bevel gear 710 can rotate independently of the second power transmitting shaft 706 about the lateral axis 712. The third bevel gear 710 can be driven by another shaft that is configured to rotatably and slidably couple with the second power transmitting shaft 706. In this way, the mechanical energy or torque input to the knuckle input gear 640 can be provided independently of the

mechanical energy or torque inputs of the first power transmitting shaft 702 and the second power transmitting shaft 706.

Referring particularly to FIG. 12, the axle 624 can include a universal joint 714. The universal joint 714 facilitates transferring the rotational kinetic energy from the driven gear 622 to the tractive element 604 and may be rotatably fixedly coupled with the shaft with which the driven gear 622 is fixedly coupled. Advantageously, the universal joint 714 facilitates providing mechanical energy to the tractive element 604 for transportation even when the tractive element 604 is steered.

Referring now to FIGS. 13-14, a suspension system 800 includes three of the suspension systems 600 that are angularly offset 120 degrees from each other. The suspension system 800 includes a housing member, a body member, a central member, etc., shown as body 802. The body 802 can be integrally formed with the outer members 608 of each of the suspension systems 600. For example, the body 802 can be integrally formed with the central member 608 of each of the suspension systems 600. The central member 608 can be co-axial with the first power transmitting shaft 702 and the second power transmitting shaft 706. The central member 608 can be rotatably coupled with a body or a frame of a vehicle and can be driven to rotate about the lateral axis 712 to thereby rotate the suspension systems 600 in unison.

Referring to FIGS. 10 and 14, the outer members 608 and the inner members 610 can define an inner volume 804. The inner volume 804 can be filled with a hydraulic fluid or any other incompressible fluid which may be selectively added or removed. In this way, the inner volumes 804 of each of the suspension systems 600 can be increased or decreased by adding fluid to the inner volumes 804. Adding fluid to the inner volumes 804 drives the inner members 610 to translate relative to the outer members 608 such that an overall longitudinal length of the suspension system 800 is increased. Removing fluid from the inner volumes 804 drives the inner members 610 to translate relative to the outer members 608 such that the overall longitudinal length of the suspension system 800 is decreased. As shown in FIGS. 13-14, the longitudinal length of the suspension system 800 can be adjusted (e.g., increased or decreased) to raise or lower the tractive element 604.

The fluid can be selectively added or removed from the inner volumes 804 to raise or lower the tractive elements 604. In some embodiments, the fluid is added or removed using a pump and can be added or removed to achieve a desired position (e.g., a desired longitudinal length) of the tractive element 604. The pump can be a hydraulic or a pneumatic pump and may be configured to provide the fluid from a reservoir. Any number of valves, connections, connectors, tubular members, etc., can be used to fluidly couple the reservoir with the inner volumes 804 of the suspension systems 600.

Extending Length Control Arm

Referring now to FIGS. 15-19, an adjustable length control arm suspension system 1000 includes a frame 1002, and tractive elements, wheels, tires, treads, etc., shown as tractive elements 1004. The tractive elements 1004 are fixedly coupled with the frame 1002 through a support assembly 1006. The support assembly 1006 can include extendable control arms, extendable/retractable frame members, telescoping support members, extendable structural members, extendable beams, bars, elongated members, etc., shown as extendable control arms 1008. The support assembly 1006 can be positioned between each of the tractive elements 1004 and the frame 1002 and can pivotally couple

with the frame 1002 and a wheel hub, a hub assembly, etc., shown as wheel hub 1013 at each tractive element 1004.

Referring particularly to FIG. 19, the extendable control arms 1008 include an outer member, a first member, an outer elongated member, an outer bar, an outer structural member, an outer support member, etc., shown as outer member 1010. The extendable control arm 1008 also includes an inner member, a second member, an inner bar, an inner structural member, an inner support member, etc., shown as inner member 1012. The inner member 1012 can be received within a cavity, a bore, a central openings, an inner volume, an internal volume, a space, etc., of the outer member 1010. The inner member 1012 can be slidably coupled with the outer member 1010 such that the inner member 1012 can translate relative to the outer member 1010. In some embodiments, outer member 1010 includes a track and the inner member 1012 is configured to slide along the track in either direction. The outer member 1010 can be a hollow bar, beam, elongated member, structural member, etc., with an internal volume that extends through at least a portion of a length of the outer member 1010 and has a cross-sectional shape that corresponds to an outer periphery of the inner member 1012.

The inner member 1012 can be driven to extend or retract relative to the outer member 1010 to increase or decrease an overall longitudinal length 1016 of the extendable control arm 1008. A longitudinal axis 1014 extends through the extendable control arm 1008 and defines a longitudinal direction along the extendable control arm 1008. The support assembly 1006 can include two sets of two of the extendable control arm 1008. The first set of the extendable control arms 1008 and the second set of the extendable control arms 1008 can be substantially parallel with each other and may be offset a distance 1022 apart between their respective longitudinal axes 1014. In some embodiments, the first set of extendable control arms 1008 are a lower set of extendable control arms 1008 that are closer to a ground surface and the second set of extendable control arms 1008 are an upper set of extendable control arms 1008 that are above the first set.

The inner member 1012 can be driven to extend or retract by an actuator 1020. The actuator 1020 can be an electric actuator, a hydraulic cylinder, etc., that is positioned within an inner volume of the outer member 1010 (e.g., the inner volume that the inner member 1012 extends into). In some embodiments, the extendable control arm 1008 itself is a cylinder. The inner volume of the outer member 1010 that the inner member 1012 extends into can be filled with a hydraulic fluid or any other incompressible fluid. Fluid can be added or removed from the inner volume to extend or retract the inner member 1012 relative to the outer member 1010. For example, fluid can be added to the inner volume to extend the inner member 1012 relative to the outer member 1010, thereby increasing the overall longitudinal length 1016 of the extendable control arm 1008. Likewise, fluid can be removed from the inner volume to retract the inner member 1012 relative to the outer member 1010, thereby decreasing the overall longitudinal length 1016 of the extendable control arm 1008.

The actuator 1020 can be positioned within the inner volume of the outer member 1010 and can include an extendable shaft. The extendable shaft can fixedly couple with an end of the inner member 1012 that extends into the inner volume of the outer member 1010. In this way, the actuator (e.g., the linear actuator) can be operated (e.g., extended or retracted) to increase or decrease the overall

longitudinal length **1016** of the extendable control arm by extending or retracting the inner member **1012** relative to the outer member **1010**.

The actuators **1020** of each of the extendable control arms **1008** can be independently operated to achieve a desired camber angle of the tractive element **1004** or to achieve a desired elevation of the frame **1002** relative to a ground surface **1024**. In some embodiments, the outer member **1010** includes a connection portion, an eyelet, an end portion, etc., shown as connection portion **1018**. The connection portion **1018** can include an aperture, an opening, a hole, etc., to facilitate coupling (e.g., pivotally coupling) the outer member **1010** with the frame **1002**. In some embodiments, the outer member **1010** pivotally or rotatably couples with the frame **1002** through the connection portion **1018**. The connection portion **1018** can be configured to receive a pin or an elongated member therethrough to rotatably or pivotally couple the extendable control arm **1008** with the frame **1002**. The connection portion **1018** can be integrally formed with an outer end (e.g., an end of the outer member **1010** that is proximate the frame **1002**) of the outer member **1010**, fixedly coupled with the outer member **1010**, etc.

The inner member **1012** can be pivotally coupled with the wheel hub **1013**. In some embodiments, the inner member **1012** includes a connection portion similar to connection portion **1018** that is configured to couple the inner member **1012** with a ball joint at the wheel hub **1013**. In this way, an angle of the extendable control arm **1008** can change, while an orientation of the frame **1002** and the tractive element **1004** may remain relatively constant (e.g., a camber angle of the tractive element **1004**).

Referring particularly to FIGS. 16-18, various operational abilities of the suspension system **1000** are shown. The extendable control arms **1008** can be extended or retracted (e.g., by operating the actuators **1020**) in unison to raise or lower the frame **1002** relative to the ground surface **1024**. For example, the actuators **1020** can be operated to extend or retract the extendable control arms **1008** a same amount, thereby elevating the frame **1002** relative to the ground surface **1024** while preserving an orientation of the frame **1002** (e.g., increasing or decreasing a distance **1028** between a bottom of the frame **1002** and the ground surface **1024**, shown in FIGS. 16-17).

The extendable control arms **1008** can also be operated to level the frame **1002**. For example, a first tractive element **1004a** may rest upon a ground surface **1024a** that is elevated relative to a ground surface **1024b** that a second tractive element **1004b** rests upon. The extendable control arms **1008** that extend between the first tractive element **1004a** and the frame **1002** may be extended to level the frame **1002**. Operating the extendable control arms **1008** that extend between the first tractive element **1004a** and the frame **1002** may increase the overall longitudinal length **1016** of the extendable control arm **1008**, thereby increasing a relative elevation between the first tractive element **1004a** and the frame **1002**. A relative elevation **1028** between central axes **1026** (e.g., central axis **1026a** of the first tractive element **1004a** and central axis **1026b** of the second tractive element **1004b**) can be controlled (e.g., increased or decreased) by operating the extendable control arms **1008**.

For example, the extendable control arms **1008** can be operated to extend, thereby increasing the distance **1028** between the frame **1002** and the ground surface **1024**. In some embodiments, if the upper set of extendable control arms **1008** and the lower set of extendable control arms **1008** are operated to extend or retract in unison (e.g., a same amount), the upper set of extendable control arms **1008** and

the lower set of extendable control arms **1008** may remain parallel, or a relative angle between the upper set of extendable control arms **1008** and the lower set of extendable control arms **1008** remains constant.

However, the upper set of extendable control arms **1008** and the lower set of extendable control arms **1008** can be operated to extend or retract different amounts to thereby change a camber angle **1030** of each tractive element **1004** as shown in FIG. 18. The camber angle **1030** is defined between a vertical axis **1032** and a central plane or a central axis **1034** of the tractive element **1004**. The lower set of extendable control arms **1008** can be extended, or the upper set of extendable control arms **1008** can be retracted (or both) to make the camber angle **1030** more negative (e.g., as shown in FIG. 18). Likewise, the lower set of extendable control arms **1008** can be retracted, or the upper set of extendable control arms **1008** can be extended (or both) to make the camber angle **1030** more positive.

Advantageously, the extendable control arms **1008** facilitate a suspension system with increased amounts of wheel travel (e.g., travel of the tractive elements **1004**), thereby facilitating avoiding interference between vehicle components and the suspension sweep. The extendable control arms **1008** also facilitate increased space availability for tire sweeps. Soft suspension rates and lower roll stiffness may be associated with increased wheel travel. By using the extendable control arms **1008**, the suspension system **1000** facilitates an adjustable track width that can improve handling stability. In some embodiments, the suspension system **1000** facilitates up to 30 inches of suspension travel and increased suspension sweep clearance. The extendable control arms **1008** can also facilitate an added degree of freedom for a transportation mode. The adjustable track width is facilitated by the extendable control arms **1008** and can improve handling stability. The extendable control arms **1008** can be actively controlled (e.g., by a control system) to maintain a desired wheel camber and to maximize wheel contact patch. For an amphibious vehicle, the suspension system **1000** facilitates providing an operation to lift and pivot the tractive elements **1004** for improved water mobility. Track width adjustments can also result in a variable motion ration, effectively changing wheel end suspension rates for different operational modes.

The control system can include a controller configured to operate the actuators **1020** to achieve a desired camber angle (e.g., a desired value of the camber angle **1030** as shown in FIG. 18) and to maintain the camber angle throughout a suspension sweep. The control system can use a variety of hardware configurations and may use a variety of control schemes to maintain the desired camber angle (e.g., feedback control, PID control, PI control, etc.).

Articulated Arm Suspension

Referring now to FIGS. 20-25, an articulated arm suspension **1100** for a vehicle includes various articulated arms **1110** and tractive elements **1104**. The tractive elements **1104** can be wheels, tires, treads, rotatable members, etc., that are configured to engage (e.g., frictionally engage) a ground surface **1124** and propel the vehicle. The vehicle can include a body, a frame, a hull, a carriage, etc., shown as body **1102**. The tractive elements **1104** are configured to engage the ground surface **1124** to transport the vehicle (e.g., to transport the body **1102**) along the ground surface **1124**. The tractive elements **1104** can be configured to support the vehicle (e.g., to support the body **1102**). Any number of tractive elements **1104** can be used (e.g., six as shown in FIGS. 20-25, four, eight, etc.).

The articulated arm suspension 1100 includes multiple articulated arms 1110 that extend between the body 1102 and the tractive elements 1104. For example, one of the articulated arms 1110 may extend between each of the tractive elements 1104 and the body 1102. The articulated arms 1110 can be configured to couple with the body 1102 at a first end (e.g., an inner end) and a wheel hub of the tractive element 1104 at an opposite end. The body 1102 defines a central longitudinal axis 1106. The articulated arms 1110 can extend at least partially in a lateral direction towards the central longitudinal axis 1106. The articulated arms 1110 can extend to the central longitudinal axis 1106 and may be coupled with the body 1102 at the central longitudinal axis 1106.

Referring particularly to FIG. 25, the articulated arm 1110 can be a telescoping member, an extendable member, a retractable member, etc. Specifically, the articulated arm 1110 can include an inner member, an inner tubular member, an extendable member, etc., shown as inner member 1114. The articulated arm 1110 can also include an outer member, an outer tubular member, a female member, etc., shown as outer member 1112. The inner member 1114 can be configured to extend or retract relative to the outer member 1112. The inner member 1114 can be received within an inner volume, a cavity, a hollow portion, etc., of the outer member 1112. The inner member 1114 and the outer member 1112 may both be tubular members and a cross-sectional shape of the inner volume or cavity of the outer member 1112 that receives the inner member 1114 may correspond to a cross-sectional shape or an outer periphery of the inner member 1114. An exterior surface (e.g., a radially outwards facing surface) of the inner member 1114 can slidably couple with an interior surface (e.g., a radially inwards facing surface of the outer member 1112). The outer member 1112 may be positioned laterally inwards relative to the inner member 1114 (e.g., closer to the central longitudinal axis 1106).

The articulated arm 1110 includes a first end, an inner end, a proximate end, etc., shown as inner end 1116 and a second end, an outer end, a distal end, etc., shown as outer end 1118. The outer end 1118 is opposite the inner end 1116. The outer end 1118 can be an outer most end of the inner member 1114 (e.g., an end of the inner member 1114 that is most proximate the corresponding tractive element 1104 or most laterally outwards). The inner end 1116 can be an inner most end of the outer member 1112 (e.g., an end of the outer member 1112 that is most proximate the central longitudinal axis 1106 or most laterally inwards).

The articulated arm 1110 is coupled, jointed, pivotally coupled, etc., with the body 1102 at the inner end 1116. The articulated arm 1110 is also coupled, jointed, pivotally coupled, etc., with a wheel hub, a hub assembly, etc., shown as wheel hub 1108 at the outer end 1118. The articulated arm 1110 can be pivotally coupled with the body 1102 at the inner end 1116 with an inner joint 1120. The inner joint 1120 can be a ball and socket joint (e.g., providing two degrees of rotational or pivotal freedom), or any other joint to provide at least one degree of rotational or pivotal freedom. In some embodiments, the inner joint 1120 has two degrees of rotational freedom. In other embodiments, the inner joint 1120 has one degree of rotational freedom (e.g., about the longitudinal axis 1106, about a lateral axis 1107 that is perpendicular to the longitudinal axis 1106, about a vertical axis 1126 (see FIGS. 21-24), or any other axis). The inner joint 1120 can be positioned at the inner end 1116 of the articulated arm 1110 (e.g., at the inner end 1116 of the outer member 1112).

The articulated arm 1110 is coupled, jointed, pivotally coupled, etc., with the wheel hub 1108 at the outer end 1118.

The articulated arm 1110 can be pivotally coupled with the wheel hub 1108 through an outer joint 1122. The outer joint 1122 can be the same as or similar to the inner joint 1120. For example, the outer joint 1122 can have a same number of degrees of rotational freedom as the inner joint 1120. In other embodiments, the outer joint 1122 has less or more degrees of rotational or pivotal freedom as the inner joint 1120. In some embodiments, the outer joint 1122 has one or more degrees of rotational or pivotal freedom about axes that are the same as the axes of the degrees of rotational/pivotal freedom as the inner joint 1120. For example, if the inner joint 1120 has a degree of rotational or pivotal freedom about the longitudinal axis 1106, the outer joint 1122 can also have a degree of rotational or pivotal freedom about the longitudinal axis 1106. In some embodiments, the outer joint 1122 has rotational/pivotal degrees of freedom that are about axes that are different than the inner joint 1120. For example, the outer joint 1122 may have a degree of freedom about an axis other than the longitudinal axis 1106 while the inner joint 1120 has a degree of freedom about the longitudinal axis 1106.

The wheel hub 1108 can be fixedly coupled with the body 1102 and may support the tractive element 1104. The tractive element 1104 can be rotatably coupled with the wheel hub 1108 such that the tractive element 1104 can rotate relative to the wheel hub 1108 to facilitate transportation of the vehicle. The tractive element 1104 can be driven by an electric motor or a rotary actuator 1129 that is coupled (e.g., fixedly) with the wheel hub 1108. In this way, the electric motor 1129 can drive the tractive element 1104 while being supported by the wheel hub 1108. In other embodiments, a drive train is routed through the articulated arms 1110. For example, the drive train may include shafts that extend through the inner volume of the articulated arms 1110 and can drive bevel gears to transfer the rotational kinetic energy to the tractive elements 1104.

The articulated arm 1110 can be selectively extended or retracted to achieve a desired overall length, and thereby achieve a desired distance or clearance (e.g., a vertical height) between the body 1102 and the ground surface 1124. The articulated arm 1110 may be a hydraulic cylinder that is filled with hydraulic fluid to increase or decrease the overall length of the articulated arm 1110 (e.g., to extend or retract the articulated arm 1110). The articulated arm 1110 can include an electric actuator, a pneumatic cylinder, etc., or any other linear actuator that is positioned within the inner volume of the outer member 1112. The actuator can be operated by a controller, a control system, a pump, a motor, an engine, etc., or any combination thereof to achieve a desired overall length of the articulated arm 1110. The actuator can be configured to drive the inner member 1114 to translate relative to the outer member 1112 (e.g., to extend from or retract into the outer member 1112).

Referring now to FIGS. 21-24, the articulated arms 1110 can be selectively operated (e.g., extended or retracted, increased or decreased in overall length, etc.) to achieve a desired clearance 1128 between the body 1102 and the ground surface 1124. The articulated arms 1110 can be operated in unison to extend or retract to increase or decrease the clearance 1128 between the body 1102 and the ground surface 1124. For example, the articulated arms 1110 can be operated to raise the body 1102 to a maximum height (shown in FIG. 24) or lowered to a minimum height (shown in FIG. 22). The articulated arms 1110 can also be operated to raise or lower the body 1102 to an intermediate height as shown in FIGS. 22 and 23.

Referring still to FIGS. 21-24, the articulated arms 1110 can be operated to pitch the body 1102 forwards 1132 or to pitch the body rearwards 1130. For example, articulated arms 1110 that are longitudinally rearward on the vehicle or the body 1102 can be operated to extend, thereby pitching the body 1102 forwards 1132. Likewise, articulated arms 1110 that are longitudinally frontward on the vehicle or the body 1102 can be operated to retract, thereby pitching the body 1102 forwards 1132. The articulated arms 1110 that are longitudinally frontward can be operated to extend, thereby pitching the body 1102 rearwards 1130. Likewise, the articulated arms 1110 that are longitudinally rearward can be operated to retract, thereby pitching the body 1102 rearwards 1130. The articulated arms 1110 that are rearward and frontward can be operated in unison (e.g., extended and retracted or vice versa) to pitch the body 1102 forwards 1132 or rearwards 1130. The amount that the articulated arms 1110 are extended or retracted can be controlled to achieve both a desired amount of pitch of the body 1102 and to achieve a desired height or clearance 1128 of the body 1102.

The articulated arms 1110 can also be operated to control or adjust a roll of the body 1102 (e.g., an angular position of the body 1102 about the central longitudinal axis 1106). For example, articulated arms 1110 that are on a right side of the body 1102 (e.g., on a right side of a plane of the body 1102 defined by the vertical axis 1126 and the central longitudinal axis 1106) may be extended or retracted to pitch the body 1102. Likewise, the articulated arms 1110 that are on a left side of the body 1102 (e.g., on a left side of a plane of the body 1102 defined by the vertical axis 1126 and the central longitudinal axis 1106) can be extended or retracted to achieve a desired pitch of the body 1102. In some embodiments, both the articulated arms 1110 on the left side of the body 1102 and the articulated arms 1110 that are on the right side of the body 1102 are operated in unison to achieve the desired pitch of the body 1102.

In this way, the pitch, roll, and clearance of the body 1102 can be controlled or adjusted by operation of the articulated arms 1110. In some embodiments, the pitch, roll, clearance/elevation, etc., of the body 1102 is maintained constant (e.g., constant over time) if a degree of extension or retraction of the articulated arms 1110 is maintained. In other embodiments, the pitch, roll, and/or clearance of the body 1102 is adjusted in real-time by operating the articulated arms 1110 in real time. For example, a control system may operate to extend or retract (e.g., independently or in unison) any of the articulated arms 1110 in real time. The control system can operate the articulated arms 1110 in real-time based on sensor feedback (e.g., based on sensor feedback indicating a degree of extension or retraction of any of the articulated arms 1110, based on accelerometers or gyroscopes that indicate an acceleration of the body 1102 in multiple directions, etc.).

Referring again to FIG. 20, the articulated arms 1110 may extend in a direction that is not completely vertical. For example, the articulated arms 1110 may extend angularly outwards from the central longitudinal axis 1106. In this way, extension or retraction of the articulated arms 1110 can increase or decrease a track width 1134. The track width 1134 is defined as a lateral distance (e.g., in the lateral direction, along the lateral axis 1107) between the tractive elements 1104. Since the articulated arms 1110 are angled outwards from the central longitudinal axis 1106, extending the articulated arms 1110 results in the track width 1134 increasing, while retracting the articulated arms 1110 results in the track width 1134 decreasing.

Advantageously, the articulated arm suspension 1100 is a suspension system with multiple degrees of freedom. The articulated arm suspension 1100 can be operated to increase or decrease the track width 1134 of the vehicle, to raise or lower the vehicle, and/or to pitch or roll the vehicle as desired. The articulated arm suspension 1100 can provide 36 inches of vertical travel. Longitudinal and lateral travel of the tractive elements 1104 can be controlled by an operator of the vehicle. The track width 1134 can be increased to facilitate improved roll resistance of the vehicle. Additionally, the articulated arms 1110 can be selectively and independently operated to raise/lower particular tractive elements 1104 to place the tractive elements 1104 on top of a step or obstacle. In this way, the articulated arms 1110 can perform a crawling operation. The vehicle may be driven forwards with a front pair of the tractive elements 1104 raised to crawl the vehicle onto the step or the obstacle. Advantageously, this can facilitate improved transportation through soft soil, mud, etc. Additionally, if one of the tractive elements 1104 becomes damaged, the articulated arms 1110 can be operated to raise the damaged tractive element 1104 so that the damaged tractive element 1104 does not contact the ground surface 1124.

Double Ball Joint Suspension System

Referring now to FIGS. 26-32, a double ball joint suspension system 1200 for a vehicle 2700 (e.g., the vehicle 10) includes a body, a hull, a carriage, etc., shown as body 1202. The body 1202 can be fixedly coupled or integrally formed with a frame member, a bar, a beam, a central frame, etc., shown as frame 1206. The frame 1206 is configured to support or couple with (e.g., indirectly) tractive elements, wheels, treads, rotational members, transportation members, etc., shown as tractive elements 1204. The tractive elements 1204 are configured to engage a ground surface 1210 and propel or transport the vehicle 2700 along the ground surface 1210.

The body 1202 can be configured to support or transport passengers, military equipment, objects, etc. For example, the body 1202 can be a carriage configured to hold and support objects or people. An underside 1215 of the body 1202 can be fixedly coupled (e.g., fastened, welded, integrally formed, etc.) with the frame 1206. The frame 1206 can be an I-beam, a U-shaped beam, etc.

The frame 1206 can define a longitudinal axis 1212 that extends centrally along the frame 1206. The longitudinal axis 1212 can define a longitudinal direction of the vehicle 2700. A linkage, a bar, a beam, an elongated member, a tie rod, a control arm, etc., shown as linkage 1208 may extend between the tractive element 1204 and the frame 1206. Specifically, the linkages 1208 can extend from the frame 1206 to a wheel hub 1214 of the tractive element 1204. A first linkage 1208a is translationally fixedly coupled with the frame 1206 at a laterally inwards end through a connection 1216. The connection 1216 can be a pivotal connection (e.g., to pivotally couple the laterally inwards end of the linkage 1208 with the frame 1206), a rotatable connection (i.e., to rotatably couple the laterally inwards end of the linkage 1208 with the frame 1206) or a ball joint (e.g., to pivotally couple the laterally inwards end of the linkage 1208 with the frame 1206 and provide several degrees of rotational freedom). In some embodiments, the connection 1216 facilitates or allows rotation of the first linkage 1208a relative to the frame 1206 in at least one direction (e.g., about an axis that extends in a direction parallel with the longitudinal axis 1212). A laterally outwards end of the first linkage 1208a that is proximate the tractive element 1204 is coupled with the wheel hub 1214 through a ball joint 1220 such that the

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wheel hub **1214** and the tractive element **1204** can rotate relative to the first linkage **1208a**. The first linkage **1208a** can extend in a direction that is laterally outwards, partially longitudinally rearwards (or forwards) and at least partially in a vertical direction (e.g., upwards or downwards). The vertical direction is substantially perpendicular to both the longitudinal axis **1212** and the lateral direction (e.g., a direction along a lateral axis **1213** as shown in FIG. 33).

A second linkage **1208b** extends between the frame **1206** and the wheel hub **1214** (e.g., between the frame **1206** and the tractive element **1204**). The second linkage **1208b** can couple with the frame **1206** at a laterally inwards end of the second linkage **1208b** through a connection **1218** at a position that is longitudinally offset relative to the connection **1216**. The connection **1218** can be a pivotal connection (e.g., to pivotally couple the laterally inwards end of the second linkage **1208b** with the frame **1206**), a rotatable connection (e.g., to rotatably couple the laterally inwards end of the second linkage **1208b** with the frame **1206**), a ball joint, etc. The connection **1218** can be longitudinally rearward of the connection **1216**.

The second linkage **1208b** extends in a direction that is at least partially laterally outwards from the longitudinal axis **1212**. The second linkage **1208b** couples with the wheel hub **1214** through the ball joint **1220**. The ball joint **1220** can be translationally fixedly coupled with the wheel hub **1214** and can pivotally couple with the laterally outwards ends of both the first linkage **1208a** and the second linkage **1208b**. In this way, the frame **1206**, the first linkage **1208a** and the second linkage **1208b** can form a triangle and may function as a control arm or an A-frame. The wheel hub **1214** and the tractive element **1204** can be driven to rotate about a steering axis **1222** that extends through the ball joint **1220**. The steering axis **1222** can be substantially vertical, depending on a camber angle of the tractive element **1204** and the wheel hub **1214**.

Referring particularly to FIGS. 26, 31-32, and 33, the suspension system **1200** can include power transmitting axles, constant velocity axles, etc., shown as axles **1234**. The axles **1234** each extend laterally outwards from the frame **1206** to the corresponding wheel hub **1214**. The axles **1234** are each configured to drive a corresponding tractive element **1204**. The axles **1234** can include a spline at a laterally outer most end that is configured to engage the tractive elements **1204** and facilitate the transfer of rotational kinetic/mechanical energy to the tractive elements **1204** for transportation. The axles **1234** can each include one or more universal joints or any other joint that is configured to facilitate the transfer of the kinetic/mechanical axis about different axes.

Referring particularly to FIGS. 26, 30, and 32-33, the suspension system **1200** includes dampers, pistons, shock absorbers, struts, etc., shown as struts **1224**, according to an exemplary embodiment. The struts **1224** can extend between the body **1202** and a corresponding wheel hub **1214**. The struts **1224** are configured to absorb shocks, impulses, or sudden forces that are transmitted to the vehicle **2700** through the tractive elements **1204** (e.g., hitting an obstacle, hitting a depression such as a pothole, etc.). The struts **1224** can be extendable/retractable members or telescoping members and may include an outer member **1226** and an inner member **1228**. The inner member **1228** can include a connection portion, an eyelet, a coupling, etc., shown as body connection portion **1230**. The body connection portion **1230** is configured to translationally fixedly couple with the body **1202** (or slidably translationally couple with the body **1202**). The body connection portion **1230** can be configured to

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pivotaly couple with the body **1202** such that the strut **1224** can pivot in at least one direction relative to the body **1202** about the body connection portion **1230**. The outer member **1226** can include a suspension connection portion, an eyelet, a connection portion, etc., shown as suspension connector **1232**. The connector **1232** can rotatably and/or pivotally couple with the wheel hub **1214** such that the wheel hub **1214** and the strut **1224** can rotate or pivot relative to each other in at least one direction (e.g., along at least one axis). The connector **1232** may facilitate translationally fixedly couple the strut **1224** with the wheel hub **1214**. In this way, the strut **1224** extends between the body **1202** and the wheel hub **1214** and is translationally fixedly coupled at either end with the body **1202** and the wheel hub **1214**. The strut **1224** may be pinned at either end with the body **1202** and the wheel hub **1214**. In other embodiments, the struts **1224** are mounted or translationally fixedly coupled at their opposite ends using ball joints, eyelets, etc. The outer member **1226** and the inner member **1228** can be rotatably fixedly coupled with each other (e.g., through a key) while being translationally coupled such that rotation of the inner member **1228** drives rotation of the outer member **1226** and vice versa.

The struts **1224** can be the same as or similar to the suspension elements described in greater detail with reference to U.S. application Ser. No. 14/684,082, now U.S. Pat. No. 9,944,145, granted Apr. 17, 2018, filed Apr. 10, 2015, the entire disclosure of which is incorporated by reference herein.

The tractive elements **1204** can be driven to rotate for steering through rotation of the struts **1224**, or through the linkages **1208**. For example, the first linkage **1208a** and/or the second linkage **1208b** can be replaced with or integrated with a linear actuator (e.g., a hydraulic actuator, an electric actuator, a pneumatic actuator, etc.). The actuator may extend or retract to drive the wheel hub **1214** and the tractive elements **1204** to rotate about their respective steering axes **1222**. In other embodiments, one of the first linkage **1208** and the second linkage **1208** is driven to translate (e.g., extend or retract relative to the frame **1206**) to drive the wheel hub **1214** and the tractive element **1204** to rotate about their respective steering axis **1222**.

Referring particularly to FIG. 33, the struts **1224** can be extended or retracted to adjust a camber angle of the tractive element **1204**. For example, the struts **1224** can retract (e.g., decrease in overall length) to produce a positive camber angle of the tractive element **1204**. Likewise, the struts **1224** can extend to rotate the tractive element **1204** to produce a negative camber angle. The struts **1224** can be operated (e.g., by a control system) by adding or removing incompressible fluid to achieve a desired angle of camber for each tractive element **1204** (e.g., independently achieve a desired camber for each tractive element **1204**).

In other embodiments, the first linkages **1208a** and the second linkages **1208b** are both replaced with or integrated with an actuator. The actuators can be operated to extend or retract the first linkages **1208a** and the second linkages **1208b** in unison, thereby adjusting the camber of the corresponding tractive element **1204**.

The struts **1224** can also be operated to extend or retract to lower or raise the tractive elements **1204**. For example, when the vehicle **2700** encounters an obstacle, a control system may operate the struts **1224** of a front set of tractive elements **1204** to retract, thereby raising the front set of tractive elements **1204** to "crawl" over the obstacle.

Referring particularly to FIGS. 27-29, the vehicle **2700** is shown performing a crawl operation up a step **2702**. In FIG. 27, the vehicle first encounters the step **2702**. In order to

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crawl up the step 2702, a control system or an operator may operate the struts 1224 to raise the front set of tractive elements 1204 (as shown in FIGS. 27-28). The operator or the control system can operate the rest of the struts 1224 to raise the tractive elements 1204 as the various sets of tractive elements 1204 engage or approach the step 2702. In this way, the vehicle 2700 can perform a crawl operation to lift itself over obstacle, steps, etc.

Referring particularly to FIG. 26, left and right tractive elements 1204 can be raised or lowered by operating the corresponding struts 1224. For example, if the vehicle 2700 is driving on uneven terrain, the struts 1224 can be operated (e.g., based on a user input or based on sensor feedback such as sensor information from a gyroscope) to level the body 1202, even if the ground surface 1210 that one of the tractive elements 1204 engages is elevated or lowered relative to the ground surface 1210 that others of the tractive elements 1204 engage. The struts 1224 can also be operated to raise or lower the body 1202, to adjust a pitch or roll angle of the body 1202, to level the body 1202, etc. For example, the struts 1224 can be operated in unison to raise or lower the body 1202 relative to the ground surface 1210, to pitch the vehicle 2700 forwards or rearwards, or to roll the body 1202 of the vehicle 2700 in either direction.

Advantageously, using the linkages 1208 facilitates a robust, lightweight suspension assembly. The suspension system 1200 also facilitates improved/adjustable ground clearance (e.g., raising and lowering the body 1202), up to 30 inches of travel (e.g., in the vertical direction) of the tractive elements 1204 (where the vertical travel of each tractive element 1204 may be each controlled independently), and reduced tire/tractive element scrub. Advantageously, the suspension system 1200 facilitates modularity such that one or more of the linkages 1208 can be easily removed and replaced with an actuator or a telescoping linkage. Replacing the linkages 1208 can also facilitate longitudinal or lateral wheel movement and an adjustable track width (e.g., an adjustable lateral distance between corresponding tractive elements 1204).

It should be understood that any of the suspension systems of FIGS. 5-33 can be used with or integrated for use with the vehicle 10. Additionally, any or all combinations of the various suspension components shown in FIGS. 5-33 should be understood to be within the scope of the present disclosure.

Configuration of Exemplary Embodiments

As utilized herein, the terms “approximately”, “about”, “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

It should be noted that the term “exemplary” as used herein to describe various embodiments is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodi-

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ments (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent, etc.) or moveable (e.g., removable, releasable, etc.). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” “between,” etc.) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

It is important to note that the construction and arrangement of the electromechanical suspension system as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements. It should be noted that the elements and/or assemblies of the components described herein may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present inventions. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred and other exemplary embodiments without departing from scope of the present disclosure or from the spirit of the appended claims.

What is claimed is:

1. A vehicle, comprising:

a chassis;

a wheel hub;

a tractive element coupled with the wheel hub; and

a suspension system, comprising:

a first extendable member coupled at a first end with the chassis and at a second end with the wheel hub; and

a second extendable member coupled at a first end with the chassis and at a second end with the wheel hub;

wherein the first extendable member and the second extendable member are configured to extend cooperatively to adjust a ride height of the vehicle, or to extend independently to adjust a camber of the tractive element, wherein, in one or more positions of extension or retraction of the first extendable member and the second extendable member, the first extendable member and the second extendable member are substantially parallel with each other.

2. The vehicle of claim 1, wherein the tractive element is rotatably coupled with the wheel hub.

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3. The vehicle of claim 1, wherein the first extendable member and the second extendable member are configured to extend or retract by operation of a plurality of actuators.

4. The vehicle of claim 1, wherein the first extendable member, the wheel hub, the second extendable member, and the chassis define a four-bar linkage.

5. The vehicle of claim 1, wherein the first extendable member and the second extendable member are configured to extend or retract to adjust a ride height of the vehicle up to 30 inches.

6. The vehicle of claim 1, wherein the first extendable member and the second extendable member are telescoping members.

7. The vehicle of claim 1, wherein the first extendable member is positioned closer to a ground surface than the second extendable member, wherein extension of the first extendable member and retraction of the second extendable member drives the camber of the tractive element to decrease.

8. A suspension assembly for a vehicle, the suspension assembly comprising:

a wheel hub and a tractive element coupled with the wheel hub;

a first extendable member coupled at a first end with a chassis and at a second end with the wheel hub; and a second extendable member coupled at a first end with the chassis and at a second end with the wheel hub;

wherein the first extendable member and the second extendable member are configured to extend cooperatively to adjust a ride height of the vehicle, or to extend independently to adjust a camber of the tractive element, wherein, in one or more positions of extension or retraction of the first extendable member and the second extendable member, the first extendable member and the second extendable member are substantially parallel with each other.

9. The suspension assembly of claim 8, wherein the first extendable member and the second extendable member are configured to extend or retract by operation of a plurality of actuators.

10. The suspension assembly of claim 8, wherein the first extendable member, the second extendable member, the wheel hub, and the chassis define a four-bar linkage.

11. The suspension assembly of claim 8, wherein the first extendable member and the second extendable member are configured to extend or retract to adjust a ride height of the vehicle up to 30 inches.

12. The suspension assembly of claim 8, wherein the first extendable member and the second extendable member are telescoping members.

13. The suspension assembly of claim 8, wherein the first extendable member is positioned closer to a ground surface than the second extendable member, wherein extension of the first extendable member and retraction of the second extendable member drives the camber of the tractive element to decrease.

14. The suspension assembly of claim 8, wherein the first extendable member and the second extendable member extend laterally outwards from the chassis.

15. A vehicle, comprising:

a chassis;

a first wheel hub and a first tractive element coupled with the first wheel hub;

a second wheel hub and a second tractive element coupled with the second wheel hub;

a first suspension assembly, comprising:

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a first extendable member coupled at a first end with the chassis and at a second end with the first wheel hub; and

a second extendable member coupled at a first end with the chassis and at a second end with the first wheel hub;

a second suspension assembly comprising:

a third extendable member coupled at a first end with the chassis and at a second end with the second wheel hub; and

a fourth extendable member coupled at a first end with the chassis and at a second end with the second wheel hub;

wherein the first extendable member and the second extendable member of the first suspension assembly are configured to extend cooperatively to adjust a ride height of the vehicle, or to extend independently to adjust a camber of the first tractive element, wherein, in one or more positions of extension or retraction of the first extendable member and the second extendable member, the first extendable member and the second extendable member are substantially parallel with each other;

wherein the third extendable member and the fourth extendable member of the second suspension assembly are configured to extend cooperatively to adjust the ride height of the vehicle or to extend independently to adjust a camber of the second tractive element; and

wherein the first suspension assembly extends from a first lateral side of the chassis and the second suspension assembly extends from a second lateral side of the chassis.

16. The vehicle of claim 15, wherein the first suspension assembly is configured to be operated independently of the second suspension assembly to raise or lower the first tractive element relative to the second tractive element.

17. The vehicle of claim 15, wherein the first extendable member and the second extendable member of the first suspension assembly are configured to extend or retract by operation of a first plurality of actuators and the third extendable member and the fourth extendable member of the second suspension assembly are configured to extend or retract by operation of a second plurality of actuators.

18. The vehicle of claim 15, wherein the first extendable member, the second extendable member, the first wheel hub, and the chassis define a first four-bar linkage, and the third extendable member, the fourth extendable member, the second wheel hub, and the chassis define a second four-bar linkage.

19. The vehicle of claim 15, wherein the first extendable member, the second extendable member, the third extendable member, and the fourth extendable member are telescoping members.

20. The vehicle of claim 15, wherein the first extendable member is positioned below the second extendable member, and the fourth extendable member is below the third extendable member, wherein extension of the first extendable member drives the camber of the first tractive element to decrease, and wherein extension of the third extendable member and retraction of the fourth extendable member drives the camber of the second tractive element to decrease.

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